Plain-language explanation: churn100-s4-c3

AI-assisted drafts for review. Nothing here has been verified by a person.

How to read this. Explanations are built up but read down: the summary first, then each part from the whole model down, in plain words, each sentence followed by the claims it rests on (C01, C02, ...; every cited claim was computed on the model and checked). Numbers are changes in predicted risk for a typical patient. The technical record (original hypotheses, formulas, evidence tables with the test check, ranges and n) is at the end. The text describes what the model does, not the world; whether it is plausible is for the reviewer to judge.

What changed in v5e (figures). v5e: every diagram is a numbered figure with a caption (list of figures after the contents) and the text refers to it; the whole-network figure has its own landscape page, scaled to the page, with a short caption and its description in the text; an input used by several parts is drawn under each of them (same input, repeated for readability, dotted outline); and an appendix figure shows the trained network before any explanation (no parts, no splits).

What changed in v5c (tables and diagrams). v5c: the evidence tables are landscape, compact and shaded by row, with the header repeated on every page (longtable and pdflscape, used in this review PDF only; the thesis preamble has neither). In the diagrams each input is drawn once, in a shared row along the bottom (the explanation's split copies of an input are listed in the technical detail instead), every cluster has one output rectangle with its plain name, the inputs that feed the output directly meet in a drawn 'sum', and a one-line legend explains the edges.

What changed in v5b (diagrams). v5b uses one grouping everywhere, the annotation hierarchy: the summary's diagram is the same picture as the whole model's section, and a part that is itself a weighted sum (blood pressure with cholesterol) is a part of its own, described as one that adds up, so the text counts three parts throughout; the lint checks every 'N parts' against the hierarchy. A full-structure diagram follows the summary: every node and connection, nested by part, with the inputs that feed the output directly in their own cluster and the unused input columns listed. Each section from the top down adds its own region of that diagram, with stubs for connections from or to other parts; each connection is drawn in exactly one section. The gate's text now says that it lowers the risk for the patients it acts on, and that more vessels and higher cholesterol take that lowering away; its vessel statements are one per group.

What changed in v5 (read down). v5 reads the explanation down. The summary (one line, a diagram, one paragraph per block) is as in v4, except that the paragraph on what adds up is split into protective items, risk-raising items and rare results, and an input that also feeds a part is quoted with its direct contribution only (the rest is in that part's paragraph). Then the document walks the annotation hierarchy from the whole model down: for each part a plain hypothesis (what it does for the prediction), plain evidence (what was checked and what it showed; for a part made of others, what they are and how they combine), a diagram one level down and where to read on. The original hypotheses, closed forms and evidence tables are in 'Technical detail' at the end. Statements about structure (what a part is made of, a copy of another part, a formula that reproduces it) are now claims of their own, checked on the graph and the data.

v4. v4 follows the structure of the explanation. The summary opens with one line and a diagram at the same grouping, then gives one paragraph per top-level block of the annotation hierarchy: one for everything that adds up independently (the inputs that feed the model score directly, and parts that are a weighted sum), and one for each part that combines its inputs in another way: what it combines, how, and for whom. 'Parts in detail' follows the hierarchy below that, with a diagram one level down for each part that contains others. The prose no longer says 'clearly', 'slightly', 'consistently' or 'on the training data only': the numbers carry it, and the test check, the ranges and n stay in the evidence tables. Only genuinely small groups are marked, with their n. The lint rejects the confidence words and checks the block structure.

Contents

1 churn100-s4-c3 **5**

1.1 Summary . . . 5
1.2 The model, read from the top down . . . 8
1.2.1 The whole model . . . 8
1.2.2 Fixed-weight inputs . . . 10
1.2.3 Service calls . . . 14
1.2.4 Early rise in calls . . . 17
1.2.5 Late rise in calls . . . 20
1.2.6 Heavy daytime use . . . 22
1.2.7 State units . . . 25
1.3 Not supported by the model . . . 27

Technical detail: churn100-s4-c3 **28**
T.1 The whole model . . . 28
T.2 Fixed-weight inputs . . . 35
T.3 Service calls . . . 40
T.4 Early rise in calls . . . 42
T.5 Late rise in calls . . . 44
T.6 Heavy daytime use . . . 46
T.7 State units . . . 48
The network before any explanation . . . 51

List of Figures

1 The model at the level of the summary: one box per part. . . . 5
2 Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. . . . 7
3 The whole model: what feeds it, one level down. . . . 8
4 The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. . . . 8
5 The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it). . . . 9
6 The whole model: Every level of state the model reads: the predicted value for the typical case at that level (dashed: the reference level); n = training customers; grey: few customers or not supported. 10
7 Fixed-weight inputs: what feeds it, one level down. . . . 11
8 Fixed-weight inputs: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. . . . 11
9 Fixed-weight inputs: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks). . . . 12
10 Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20). . . . 12
11 Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20). . . . 13
12 Fixed-weight inputs: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). . . . 14
13 Service calls: what feeds it, one level down. . . . 14
14 Service calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. . . . 14
15 Service calls: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks). . . . 15
16 Service calls: How the effect differs between groups: one curve per group, for the typical case of each group. . . . 15
17 Service calls: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20). . . . 16
18 Service calls: This part's own output, unit h23 (combines 3 non-linear units), on its own scale (its activation), with every other input at the typical case; it is active for 831 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data. 16
19 Service calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). . . . 17
20 Early rise in calls: what feeds it, one level down. . . . 17
21 Early rise in calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. . . . 17
22 Early rise in calls: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks). . . . 18

23 Early rise in calls: How the effect differs between groups: one curve per group, for the typical case of each group. . . . 18
24 Early rise in calls: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20). . . . 18
25 Early rise in calls: This part's own output, unit h11 (the output of 'Early rise in calls'), on its own scale (its activation), with every other input at the typical case; it is active for 830 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data. . . . 19
26 Early rise in calls: This part's own output, unit h9 (the output of 'Early rise in calls'), on its own scale (its activation), with every other input at the typical case; it is active for 830 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data. . . . 19
27 Early rise in calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). . . . 20
28 Late rise in calls: what feeds it, one level down. . . . 20
29 Late rise in calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. . . . 20
30 Late rise in calls: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks). . . . 21
31 Late rise in calls: How the effect differs between groups: one curve per group, for the typical case of each group. . . . 21
32 Late rise in calls: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20). . . . 21
33 Late rise in calls: This part's own output, unit h6 (combines 2 non-linear units), on its own scale (its activation), with every other input at the typical case; it is active for 134 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data. . . . 22
34 Late rise in calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). . . . 22
35 Heavy daytime use: what feeds it, one level down. . . . 23
36 Heavy daytime use: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. . . . 23
37 Heavy daytime use: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks). . . . 23
38 Heavy daytime use: How the effect differs between groups: one curve per group, for the typical case of each group. . . . 24
39 Heavy daytime use: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20). . . . 24
40 Heavy daytime use: This part's own output, unit h13 (the output of 'Heavy daytime use'), on its own scale (its activation), with every other input at the typical case; it is active for 929 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data. . . . 24
41 Heavy daytime use: This part's own output, unit h5 (the output of 'Heavy daytime use'), on its own scale (its activation), with every other input at the typical case; it is active for 818 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data. . . . 25
42 Heavy daytime use: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). . . . 25
43 State units: what feeds it, one level down. . . . 25
44 State units: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. . . . 26
45 State units: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20). . . . 26
46 State units: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom). . . . 27

47 The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): Alabama, Iowa, Idaho, Kentucky, Arkansas, Maryland, Michigan, Minnesota, Missouri, New Hampshire, New Jersey, New Mexico, Ohio, South Dakota, Texas, Wisconsin, West Virginia, Wyoming, Connecticut, Washington DC, area code 408, area code 510, total evening minutes, total evening calls, total night-time minutes, total night-time calls, total night-time charge, total international minutes, total international calls, total international charge. 52

1 churn100-s4-c3

Genome `27f540ac-6acb-4d58-9972-cd71fae31cf0`, explanation `ffdb207f-1e0f-4d2e-a330-f47b1ca75c4a`. Split: train primary (n=4000), test as a check (n=1000). Claims: 123, verified 121, not supported 2.
AI-assisted draft for review (unverified)

Reference (typical) customer: the typical customer: the median training customer for every number and the most common training category for every category (not a real customer). account length: 100; area code: area code 415; international plan: no international plan; number customer service calls: 1 calls; number vmail messages: 0 messages; state: West Virginia; total day calls: 100 calls; total day charge: 30.69 US dollars; total day minutes: 180 minutes; total eve calls: 100 calls; total eve charge: 17.12 US dollars; total eve minutes: 201 minutes; total intl calls: 4 calls; total intl charge: 2.78 US dollars; total intl minutes: 10 minutes; total night calls: 100 calls; total night charge: 9.02 US dollars; total night minutes: 200 minutes; voice mail plan: no voice-mail plan. Predicted chance that a customer leaves: 1%.

1.1 Summary

For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, no voice-mail messages, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up four parts: fixed-weight inputs, service calls, heavy daytime use and state units. [C054] Figure 1 shows the model at this level: one box per paragraph below.

How to read the diagrams. Edge colour is the sign of that one connection: blue positive, red negative; thicker is a larger weight. The outline and arrow on an input (and on a part's output box) is its overall effect on the final prediction, read from the verified claims: orange and up arrow raises it, teal and down arrow lowers it, grey and double arrow depends (it goes both ways, or differs between levels). A negative connection into a part that itself raises the prediction means that input lowers the prediction; through a hinge the overall effect can only be read from the claims, not from the edges. Up arrow means towards churn; down arrow, away from it. A double outline marks an input (or a part's output) that acts at a fixed rate everywhere: it adds the same amount to the score per unit, so it is explained by its rate alone (see 'Linear factors'); a dashed double outline, at a fixed rate on the observed data. The normal outline is an input whose effect switches on and off somewhere. For example, a voice-mail plan has a negative connection into 'Fixed-weight inputs', a part that raises the predicted chance that a customer leaves, so a voice-mail plan lowers it.

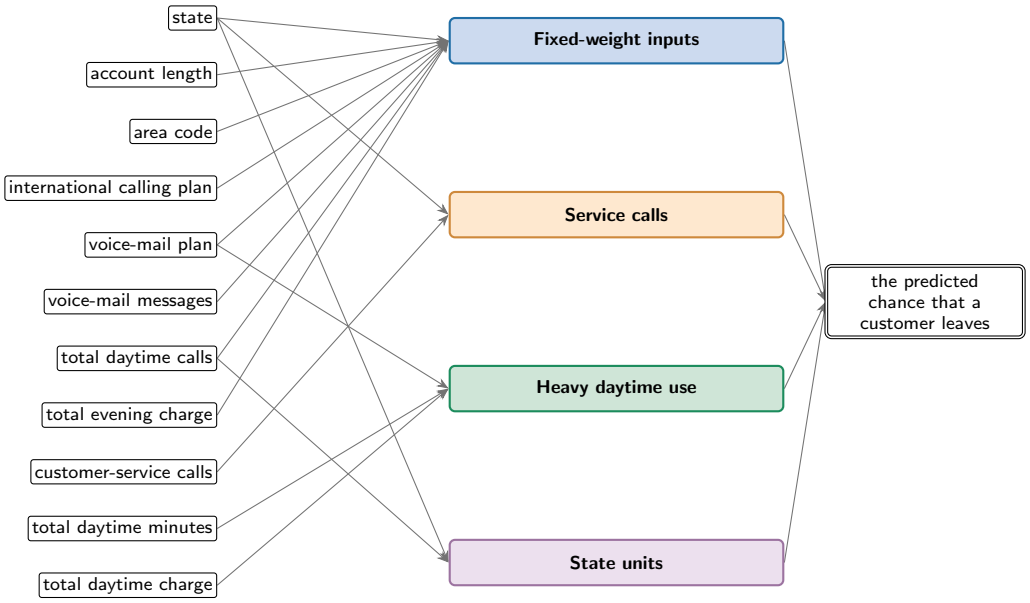


Figure 1: The model at the level of the summary: one box per part.

Fixed-weight inputs. This part adds up most inputs with fixed weights: an international plan raises a typical customer's chance of leaving from about 1% to about 18%, and a voice-mail plan lowers it to below 1%. [C055, C063] More voice-mail messages raise it steadily, from about 1% at 0 to about 6% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C056] A higher evening charge also raises it steadily, from about 1% at 11.76 US dollars to about 2% at 22.57 US dollars, flattening out near the bottom because the predicted chance cannot go much lower. [C059] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and about 4% in Nevada, and to below 1% in Rhode Island, Hawaii and Vermont (72 to 82 training customers per state). [C057, C058, C061, C062, C064]

Service calls. This part adds up two parts: an early rise and a late rise in calls to customer service. [C090] From about 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls take it from about 1% to about 10%, flattening out near the bottom because the predicted chance cannot go much lower. [C089] Up to 3 calls the chance rises slowly at first, then faster, from about 1% to about 2% for a typical customer, flattening out near the bottom because the predicted chance cannot go much lower. [C084] That early rise happens only outside Nebraska, California and Colorado: in Nebraska the chance stays at about 1% from 0 to 3 calls, against a rise from about 1% to about 2% elsewhere, slowly at first, then faster, flattening out near the bottom because the predicted chance cannot go much lower. [C088]

Heavy daytime use. Daytime minutes add to the chance only once they rise above about 209 minutes, in an uneven way: for a typical such customer, from about 1% at 0 minutes to about 16% at 271 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C100] The rise is much larger without a voice-mail plan: from 112 to 249 daytime minutes the chance goes from about 1% to about 6% without one, but stays below 1% with one, in an uneven way, flattening out near the bottom because the predicted chance cannot go much lower. [C099] This part acts on 1074 of 4000 training customers, 290 of whom left; for the typical customer it acts on, the chance is about 1% without it and about 4% with it. [C102]

State units. This part reads only the state: it raises a typical customer's chance from about 1% to about 12% in Rhode Island, to about 3% in Vermont and to about 2% in Tennessee. [C104, C105, C106] It lowers the chance to below 1% in Virginia and in Nevada. [C107, C108]

The whole network. Figure 2 (next page, landscape) draws every unit and connection of the explained model, nested by part: shaded clusters are the parts of the model and darker boxes the parts inside them; circles are hidden units (h1, h2, . . .); each cluster's rectangle is its output, and a dashed 'sum' stands for an addition the next unit does (drawn, not a unit of the model); the hexagon is the model's output. An input used by several parts is drawn under each of them (same label, dotted outline, the first copy marked with the number of copies): it is the same input, repeated for readability. Not used by the model (input columns with no connection): state: 20 of 51 levels, area code 408, area code 510, total evening minutes, total evening calls, total night-time minutes, total night-time calls, total night-time charge, total international minutes, total international calls, total international charge.

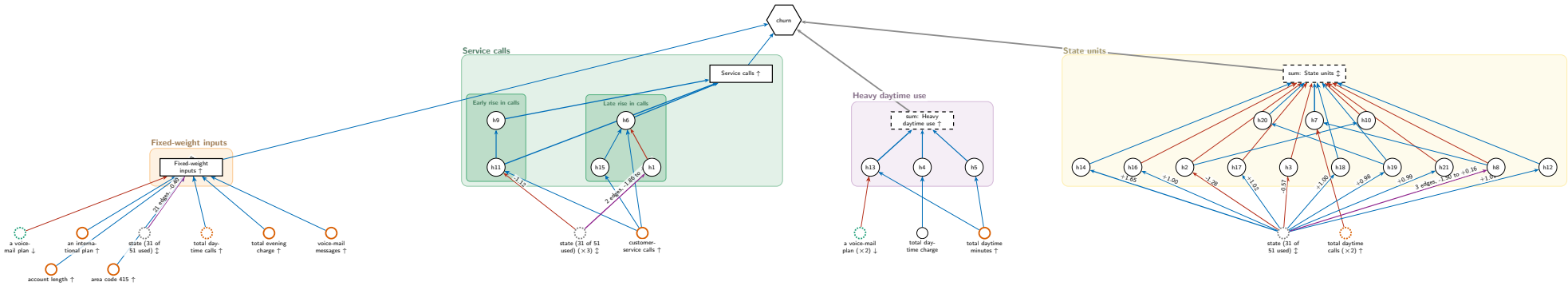


Figure 2: Every unit and connection of the explained model, nested by part. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Worth checking against what is known. Worth checking: for customers in Rhode Island the fixed-weight inputs lower the chance to below 1% while the state units raise it to about 12%, so overall it is about 1%. [C061, C104, C025] Worth checking: for customers in Nevada the fixed-weight inputs raise the chance to about 4% while the state units lower it to below 1%, so overall it is about 2%. [C058, C108, C018] Worth checking: the state moves the chance by several points (Washington about 5%, Maine about 3%, Hawaii below 1%), based on 72 to 86 training customers per state. [C004, C006, C007] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C008, C002] Worth checking: the early rise with customer-service calls does not happen for customers in Nebraska. [C093]

1.2 The model, read from the top down

Each part of the explanation in turn, from the whole model down to the smallest parts: what it does for the prediction (hypothesis), what was checked (evidence), a diagram one level down, and where to read on. The technical record of each part is in "Technical detail" at the end.

1.2.1 The whole model

Figure 3 shows what feeds this part; Figure 4 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

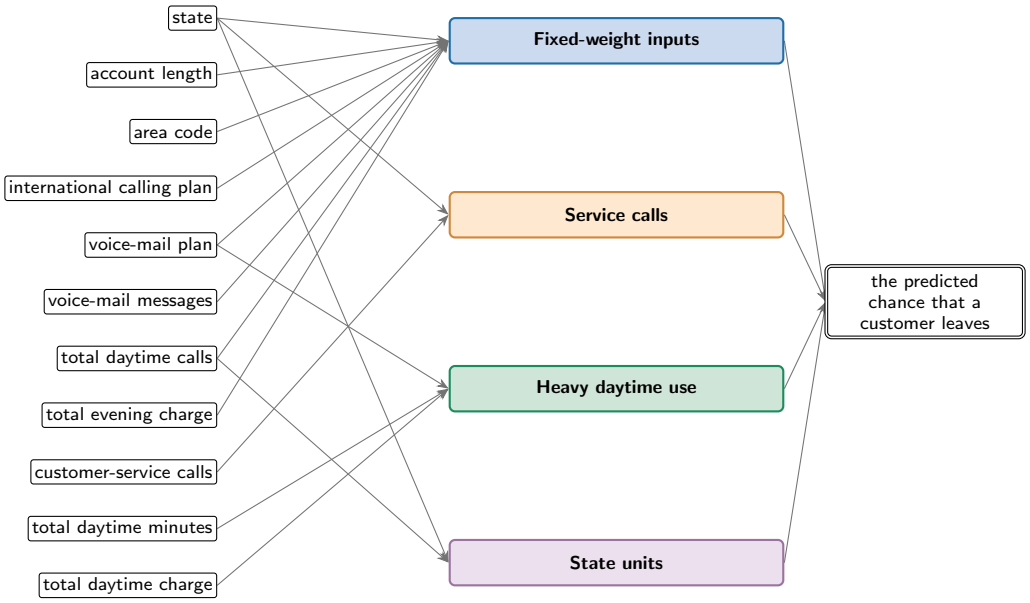


Figure 3: The whole model: what feeds it, one level down.

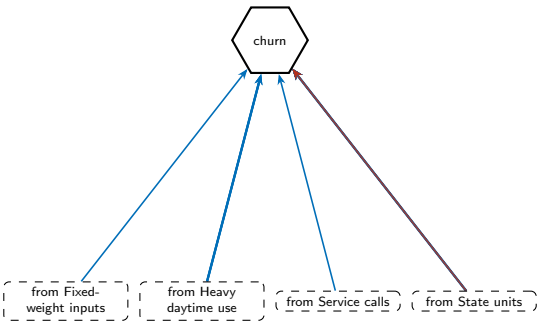


Figure 4: The whole model: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. The model adds up four parts into one score: fixed-weight inputs, service calls, heavy daytime use and state units. [C054]

Evidence. It is made up of these four parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C054] Changing one input at a time across the whole model, an international

plan takes a typical customer's chance of leaving from about 1% to about 18%, and a voice-mail plan lowers it to below 1%. [C001, C008] More voice-mail messages raise it steadily, from about 1% at 0 to about 6% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C002] More daytime minutes raise it in an uneven way, from about 1% at 112 minutes to about 6% at 249 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C003] The model ignores evening calls, evening minutes, international calls, the international charge, international minutes, night-time calls, the night-time charge and night-time minutes. [C046, C047, C048, C049, C050, C051, C052, C053]
Evidence plots: Figure 5, Figure 6.

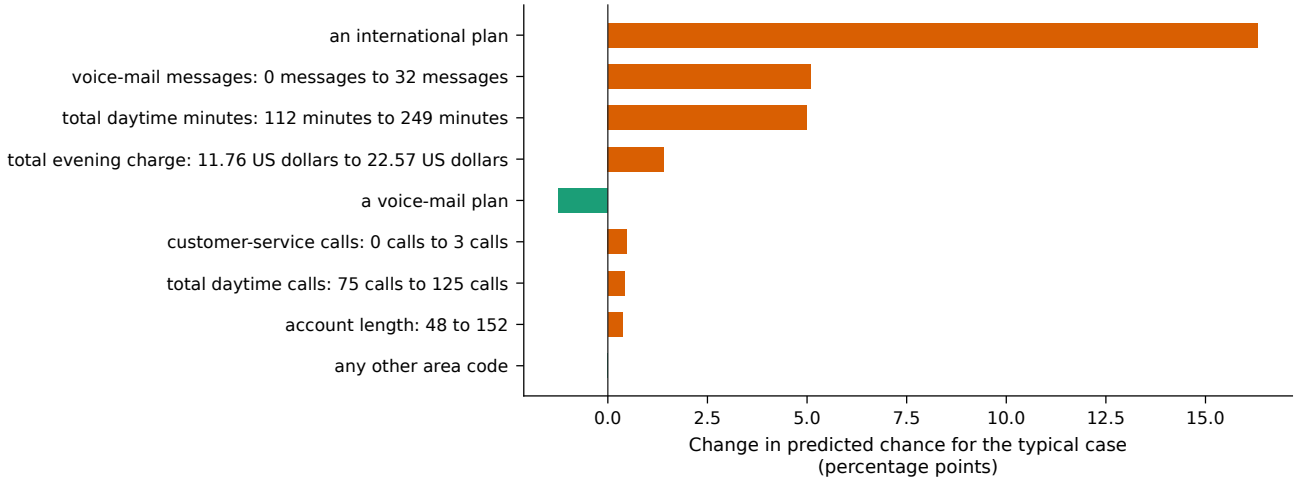


Figure 5: The whole model: Each input's change in the predicted value for the typical case across the whole model, largest first (orange: raises it, teal: lowers it).

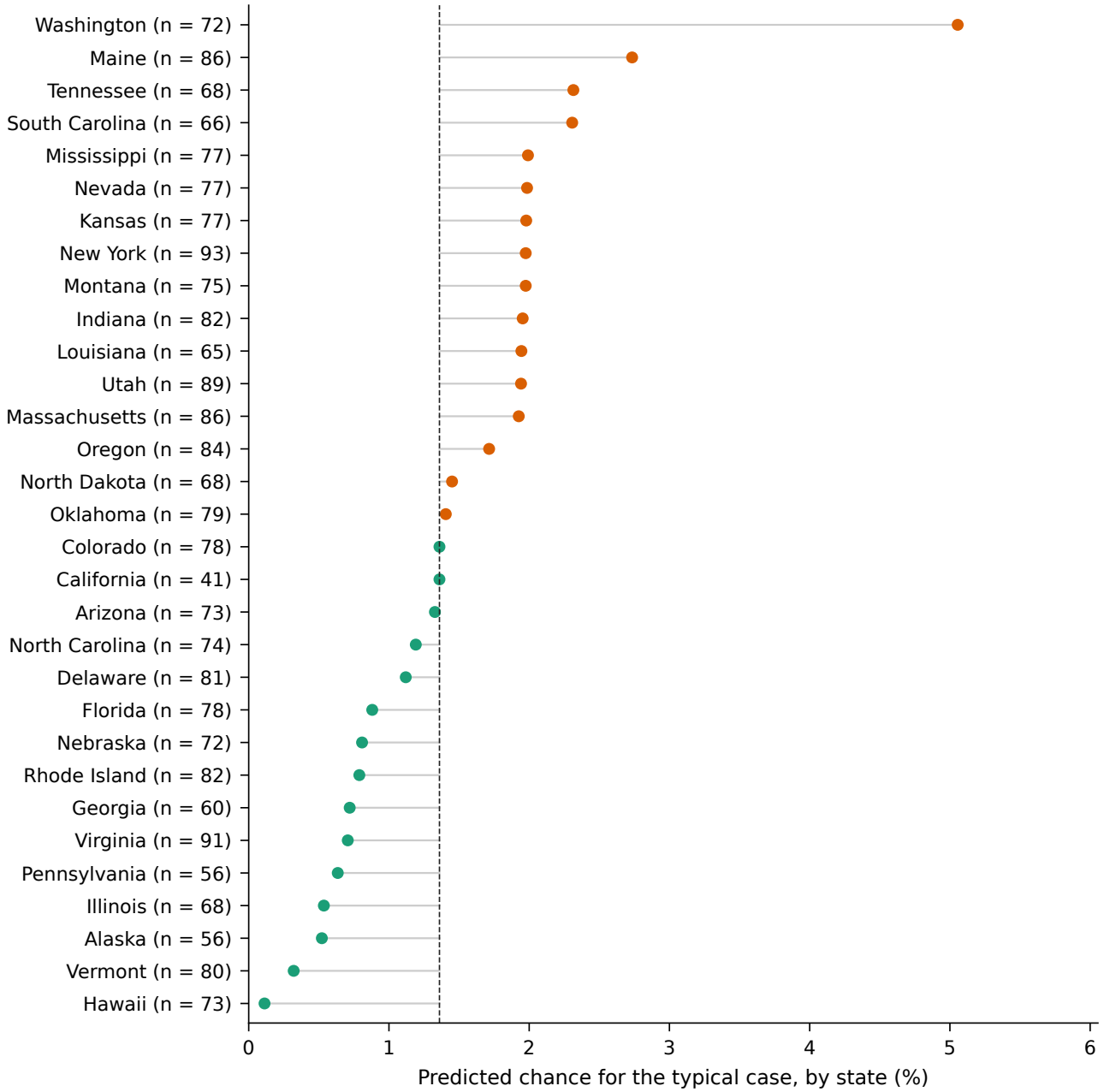


Figure 6: The whole model: Every level of state the model reads: the predicted value for the typical case at that level (dashed: the reference level); n = training customers; grey: few customers or not supported.

Dive deeper: Fixed-weight inputs (1.2.2); Service calls (1.2.3); Heavy daytime use (1.2.6); State units (1.2.7). Technical detail: T.1.

1.2.2 Fixed-weight inputs

Part of: The whole model (1.2.1).
Figure 7 shows what feeds this part; Figure 8 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

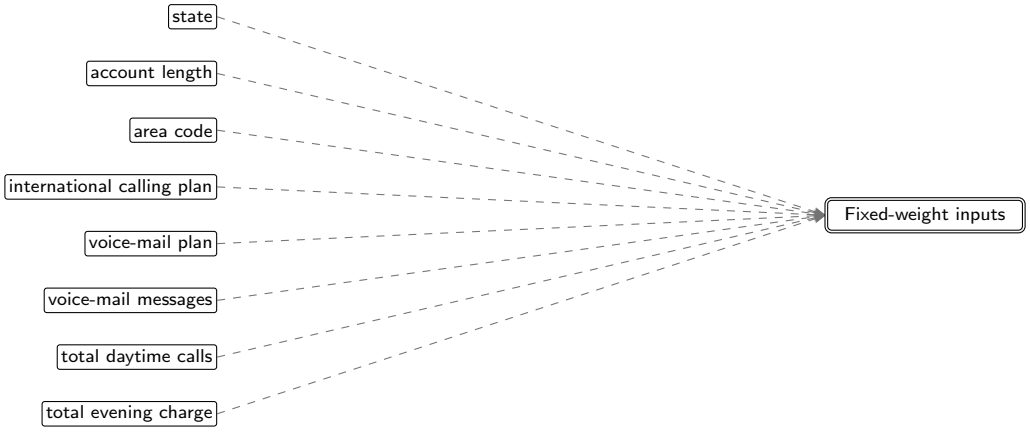


Figure 7: Fixed-weight inputs: what feeds it, one level down.

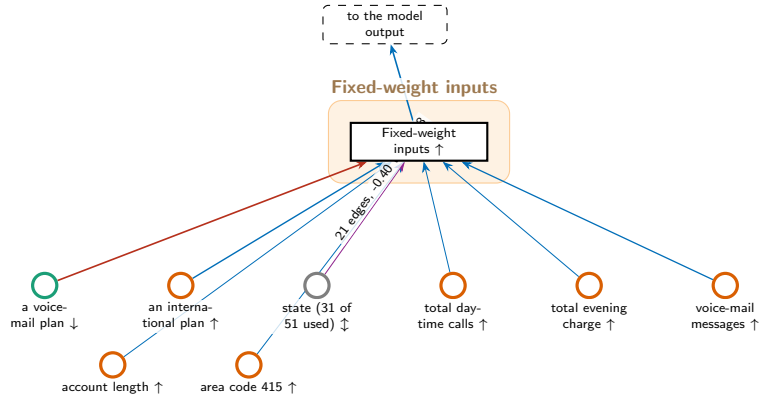


Figure 8: Fixed-weight inputs: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part adds up most inputs with fixed weights: an international plan, voice-mail messages and the evening charge raise the chance of leaving, steadily for the numbers, and a voice-mail plan lowers it. [C055, C056, C059, C063]

Evidence. Through it, an international plan takes a typical customer from about 1% to about 18%, and a voice-mail plan lowers the chance to below 1%. [C055, C063] More voice-mail messages raise it steadily, from about 1% at 0 to about 6% at 32 messages, flattening out near the bottom because the predicted chance cannot go much lower. [C056] A higher evening charge also raises it steadily, from about 1% at 11.76 US dollars to about 2% at 22.57 US dollars, flattening out near the bottom because the predicted chance cannot go much lower. [C059] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and about 4% in Nevada, and to below 1% in Rhode Island, Hawaii and Vermont (72 to 82 training customers per state). [C057, C058, C061, C062, C064] Its formula reproduces the part exactly on all 4000 training customers. [C083]
Evidence plots: Figure 9, Figure 10, Figure 11, Figure 12.

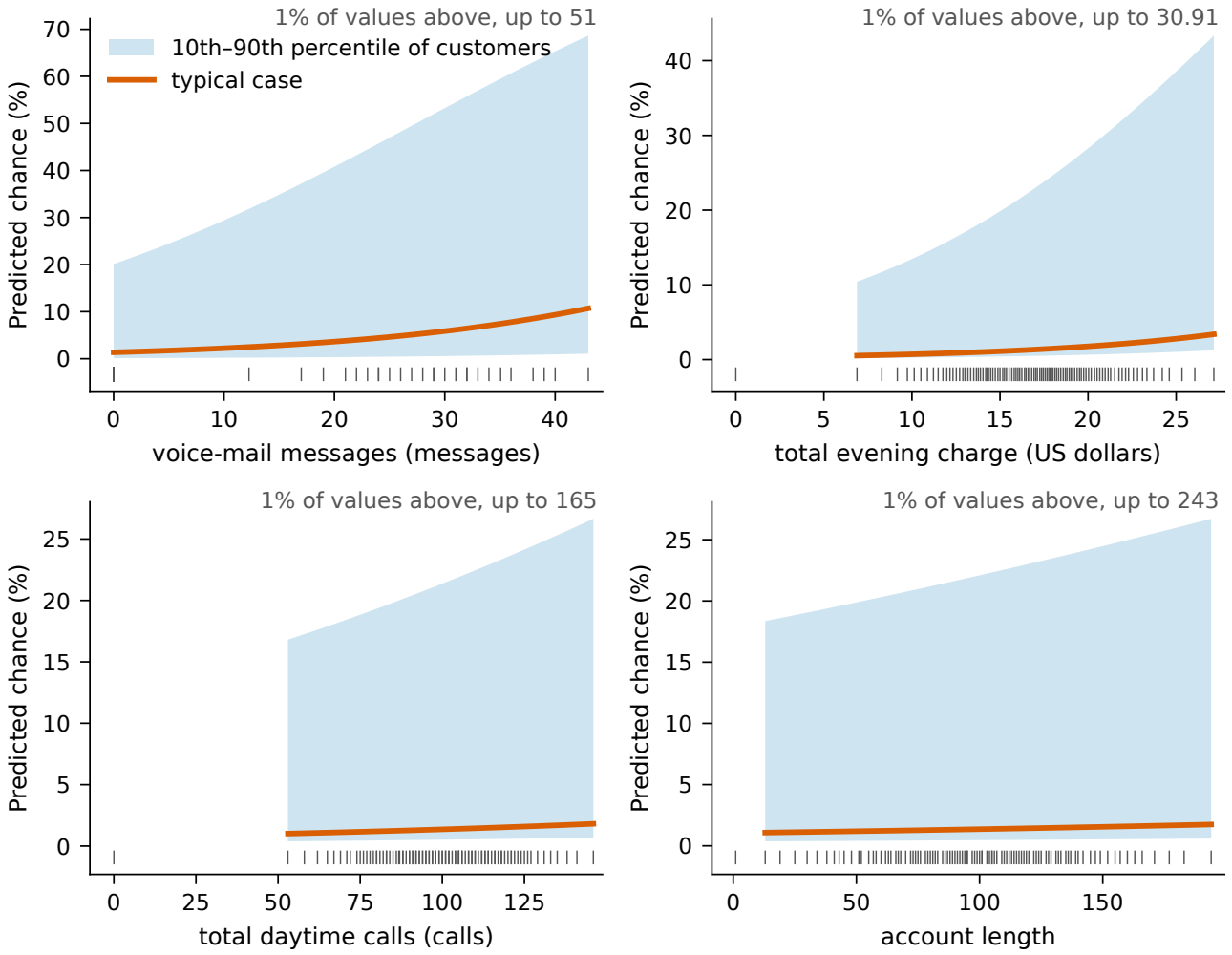


Figure 9: Fixed-weight inputs: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

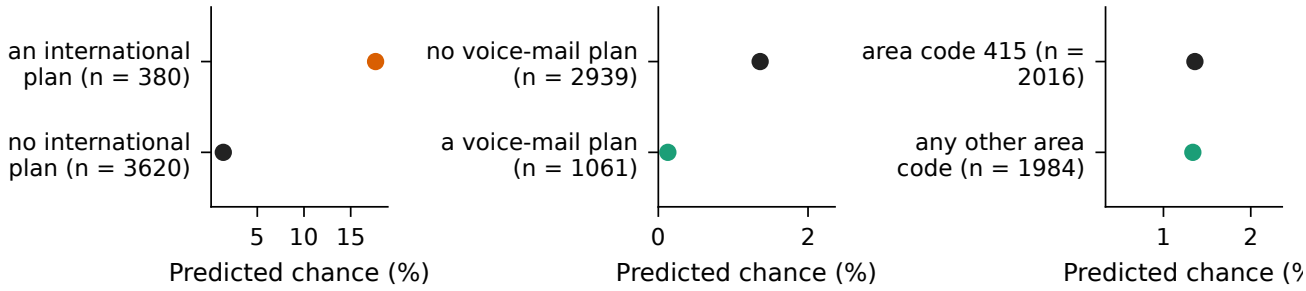


Figure 10: Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

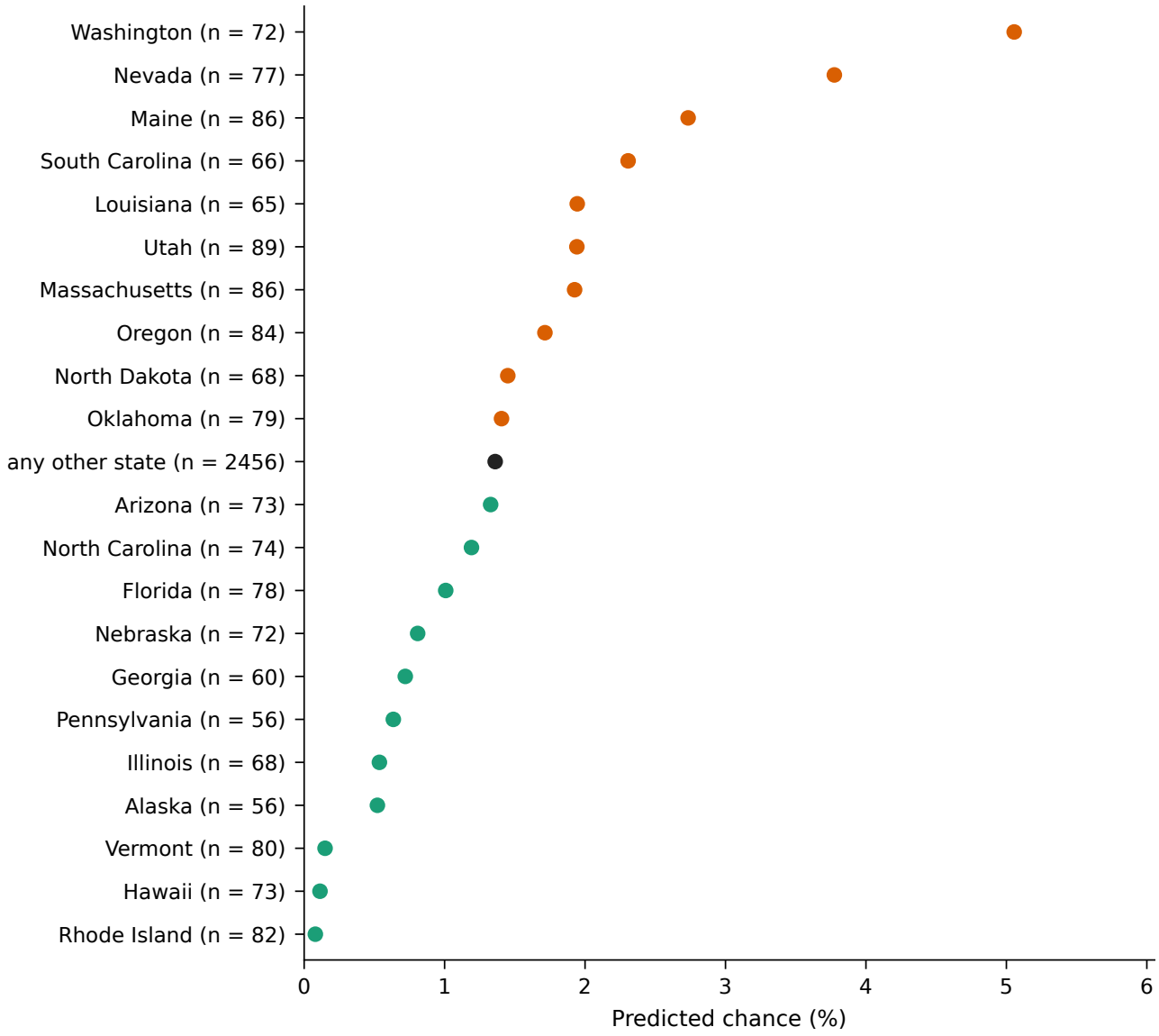


Figure 11: Fixed-weight inputs: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

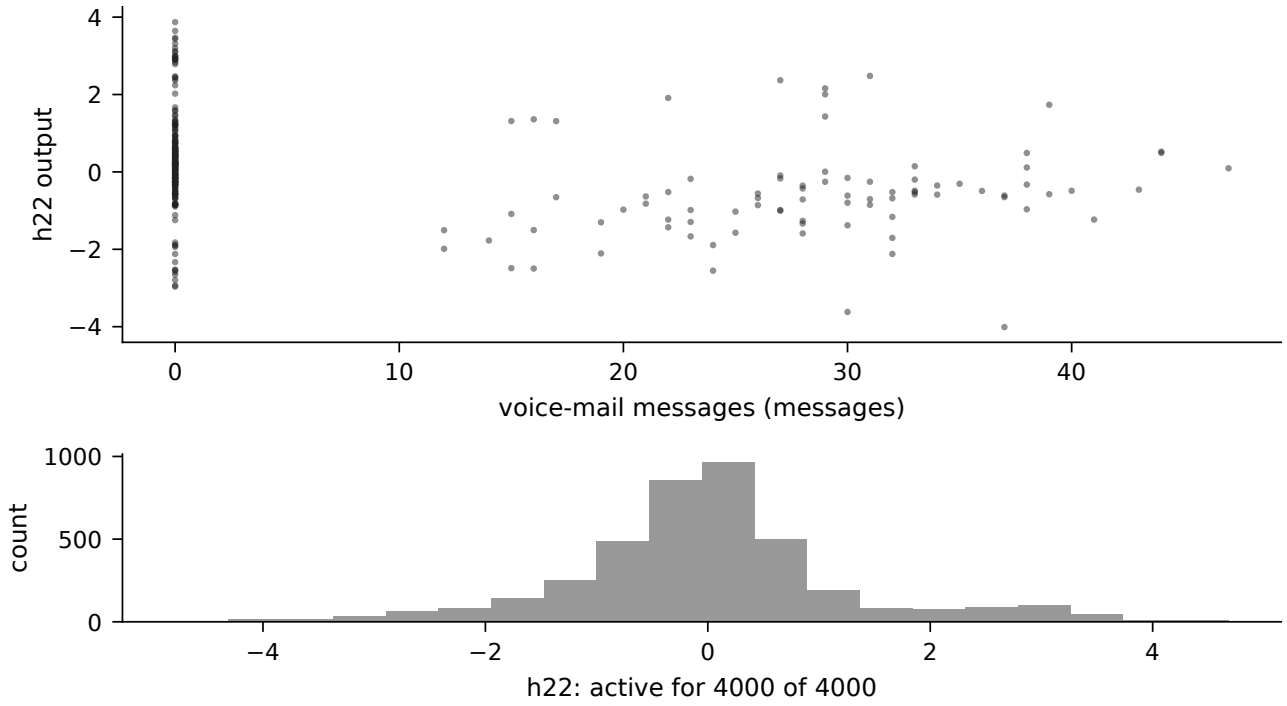


Figure 12: Fixed-weight inputs: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.2.

1.2.3 Service calls

Part of: The whole model (1.2.1).
Figure 13 shows what feeds this part; Figure 14 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

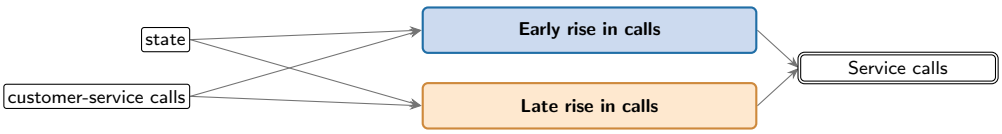


Figure 13: Service calls: what feeds it, one level down.

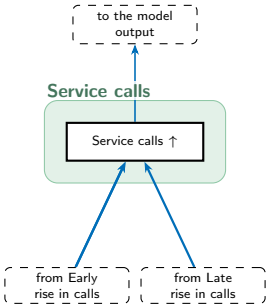


Figure 14: Service calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving with calls to customer service, slowly at first, then faster from about 3 calls upwards; it adds up two parts. [C090, C089]

Evidence. It is made of two parts that are added: an early rise and a late rise in calls to customer service. [C090] From about 3 calls upwards it adds to the chance slowly at first, then faster: for a typical such customer, 4 calls

take it from about 1% to about 10%, flattening out near the bottom because the predicted chance cannot go much lower. [C089] Up to 3 calls the chance rises slowly at first, then faster, from about 1% to about 2% for a typical customer, flattening out near the bottom because the predicted chance cannot go much lower. [C084] That early rise happens only outside Nebraska, California and Colorado: in Nebraska the chance stays at about 1% from 0 to 3 calls, against a rise from about 1% to about 2% elsewhere, slowly at first, then faster, flattening out near the bottom because the predicted chance cannot go much lower. [C088]
Evidence plots: Figure 15, Figure 16, Figure 17, Figure 18, Figure 19.
Figure 18 shows this part's own output, unit h23: it switches on at about 2.827 calls of customer-service calls, and is zero below.

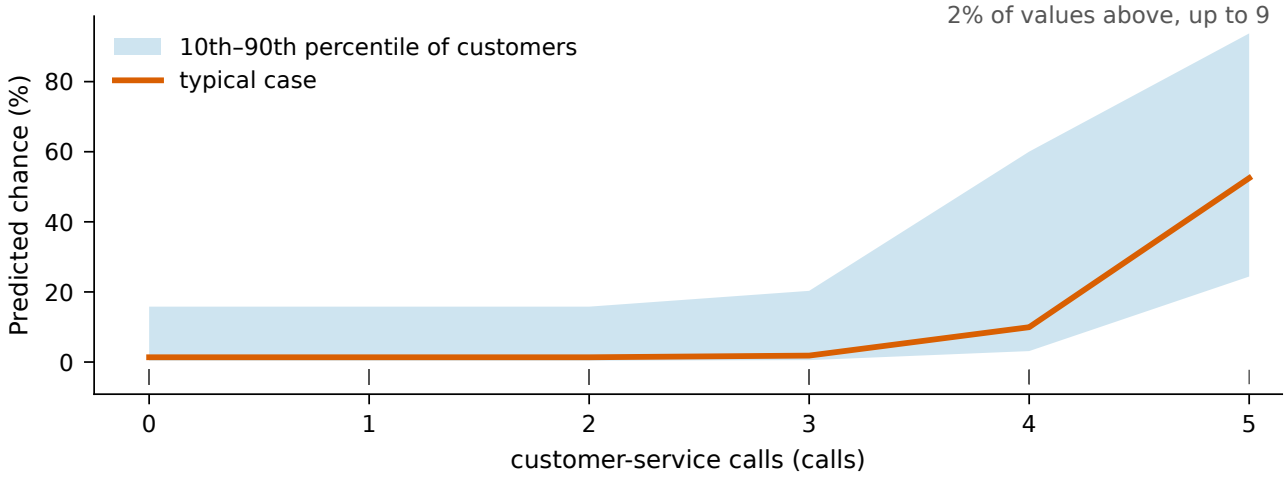


Figure 15: Service calls: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

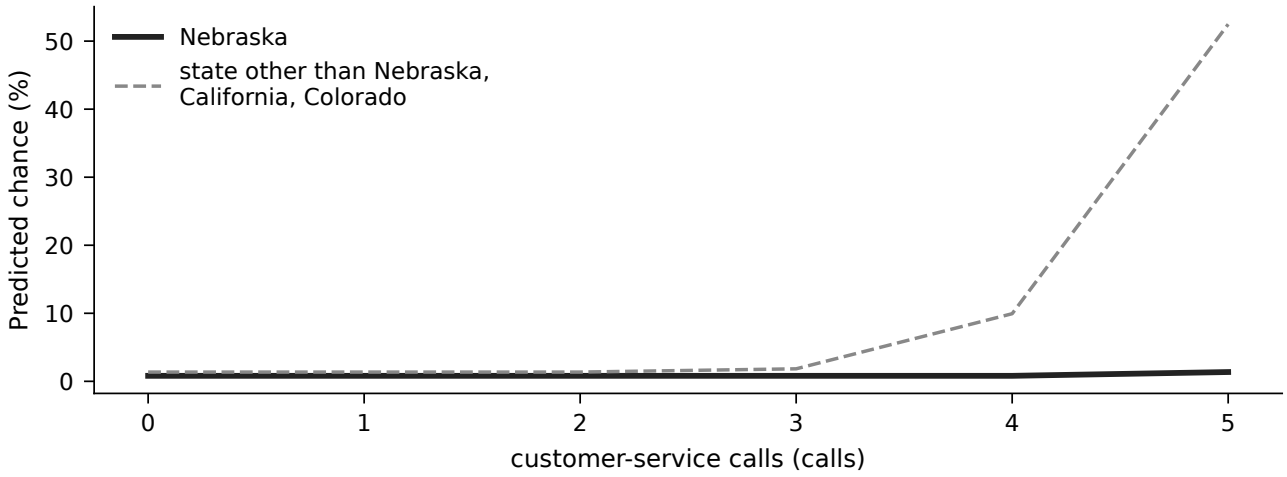


Figure 16: Service calls: How the effect differs between groups: one curve per group, for the typical case of each group.

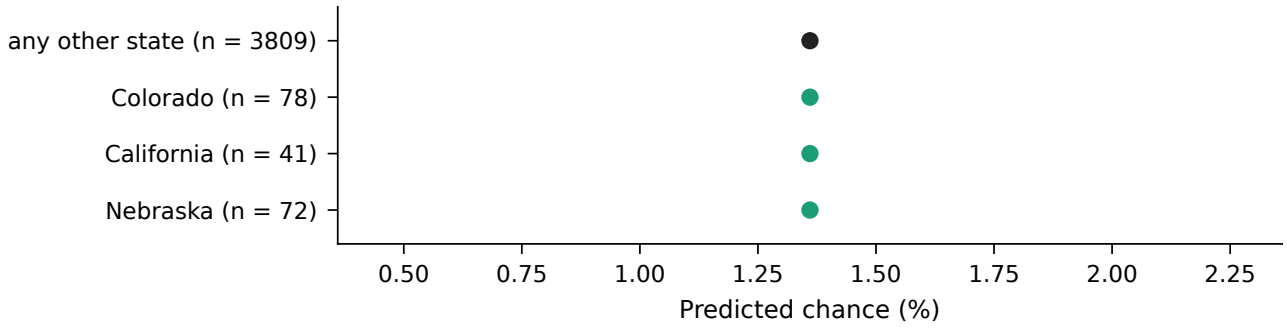


Figure 17: Service calls: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

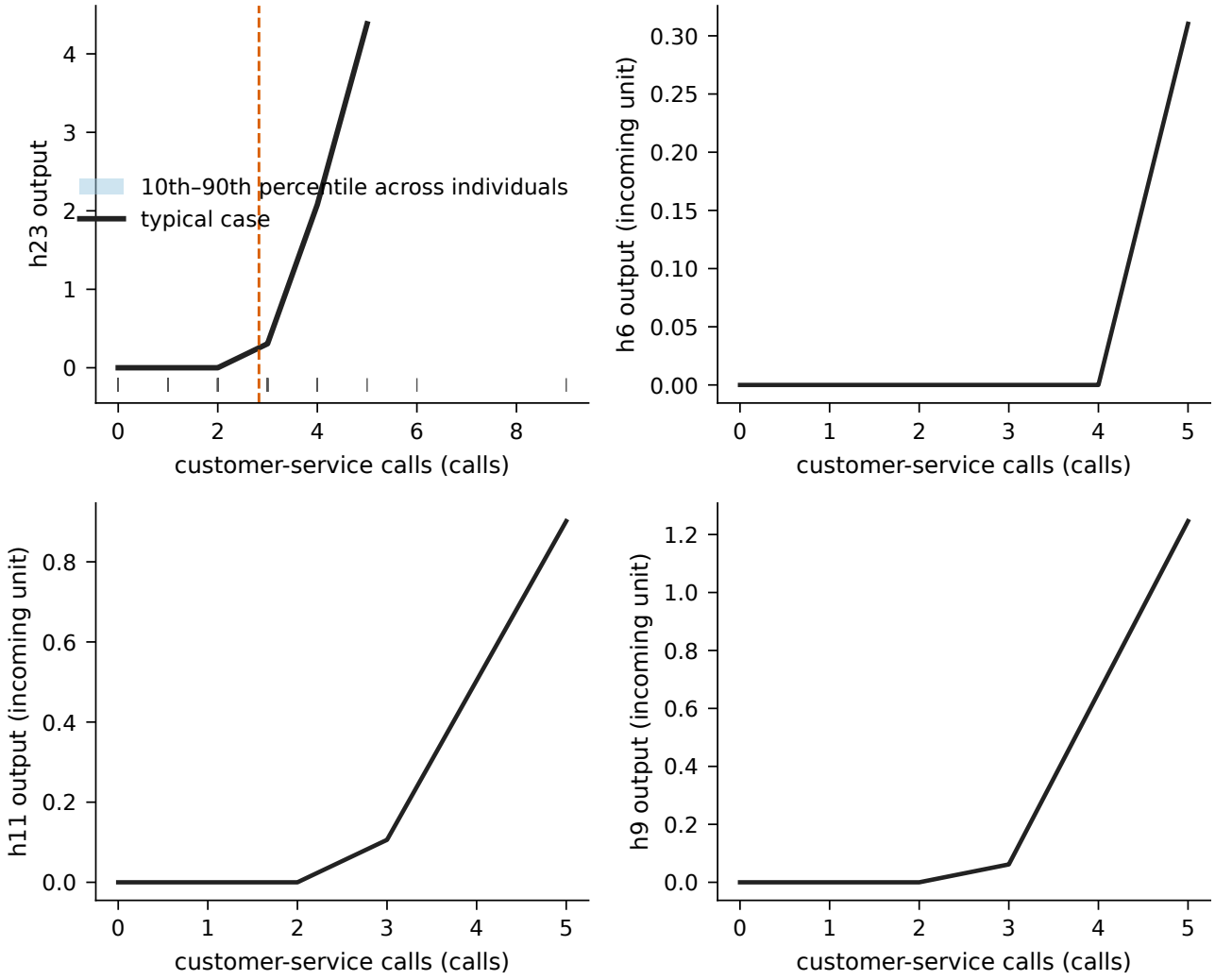


Figure 18: Service calls: This part's own output, unit h23 (combines 3 non-linear units), on its own scale (its activation), with every other input at the typical case; it is active for 831 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

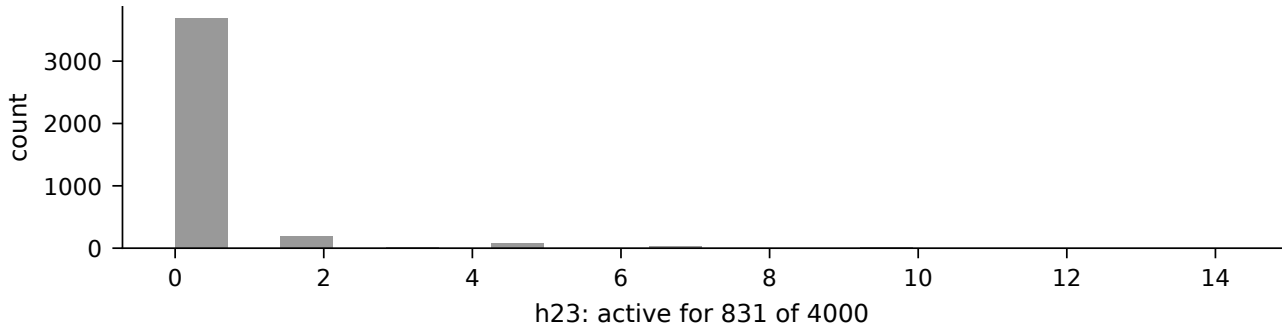


Figure 19: Service calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Dive deeper: Early rise in calls (1.2.4); Late rise in calls (1.2.5). Technical detail: T.3.

1.2.4 Early rise in calls

Part of: Service calls (1.2.3).
Figure 20 shows what feeds this part; Figure 21 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

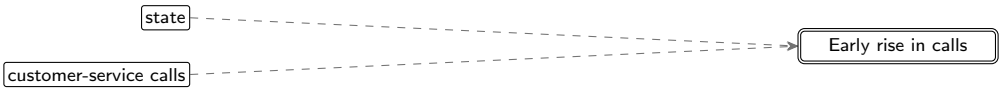


Figure 20: Early rise in calls: what feeds it, one level down.

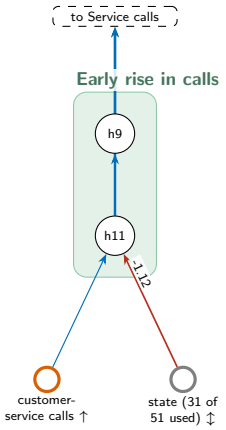


Figure 21: Early rise in calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving from about 3 customer-service calls upwards, in an uneven way. [C094]

Evidence. It acts on 830 of 4000 training customers, 213 of whom left; for the typical customer it acts on, the chance is about 1% without it and about 2% with it. [C095] Through it, calls add to the chance only from about 3 calls upwards, in an uneven way: for a typical such customer, 4 calls take it from about 1% to about 10%, flattening out near the bottom because the predicted chance cannot go much lower. [C094] Up to 3 calls it raises the chance slowly at first, then faster, but only outside Nebraska: there from about 1% to about 2%, while in Nebraska it stays at about 1%, flattening out near the bottom because the predicted chance cannot go much lower. [C093] Its formula reproduces the part exactly on all 4000 training customers. [C096]
Evidence plots: Figure 22, Figure 23, Figure 24, Figure 25, Figure 26, Figure 27.
Figure 25 shows this part's own output, unit h11: it switches on at about 2.734 calls of customer-service calls, and is zero below.
Figure 26 shows this part's own output, unit h9: it switches on at about 2.896 calls of customer-service calls, and is zero below.

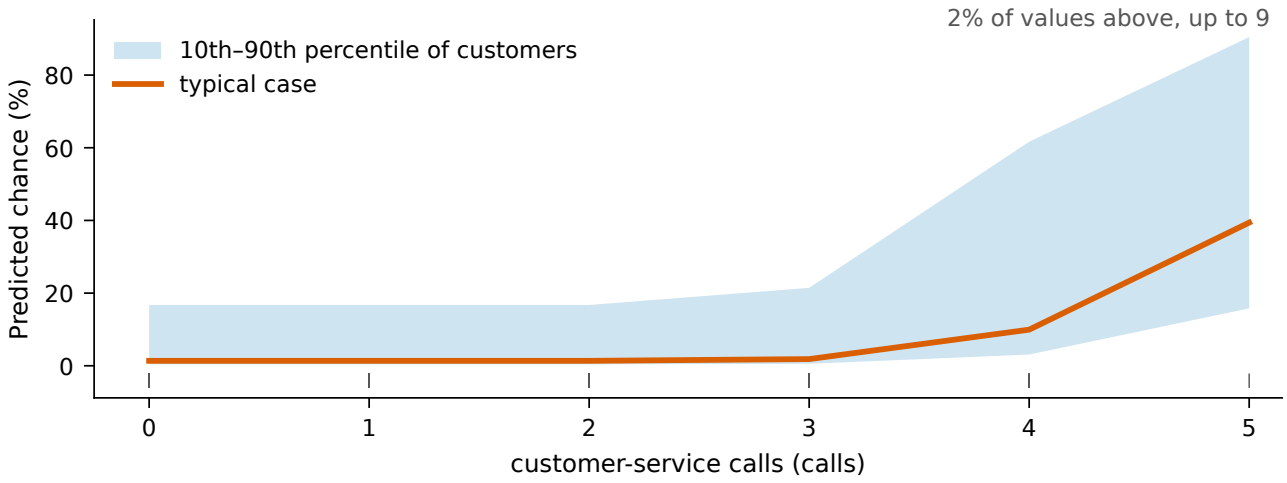


Figure 22: Early rise in calls: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

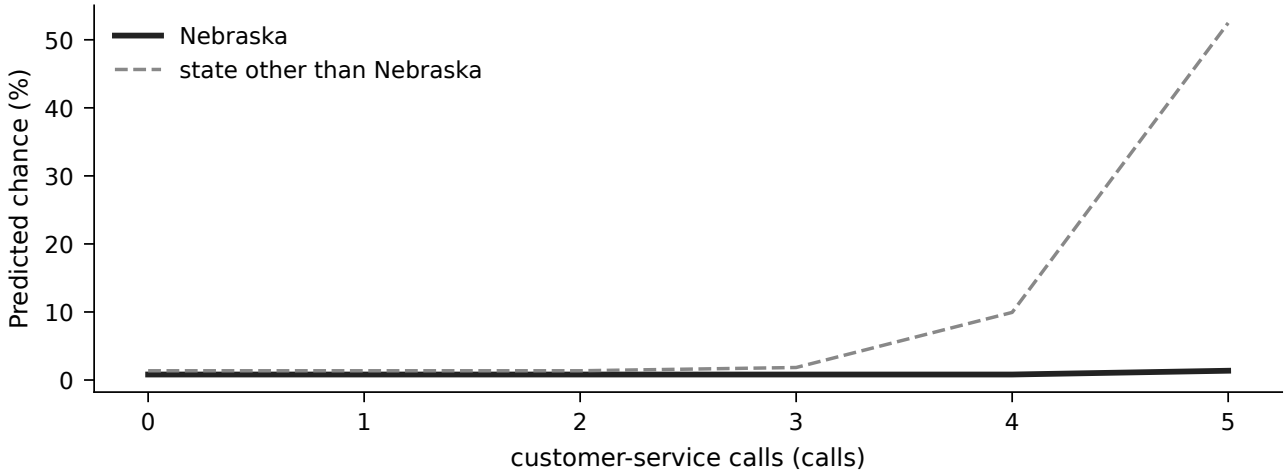


Figure 23: Early rise in calls: How the effect differs between groups: one curve per group, for the typical case of each group.

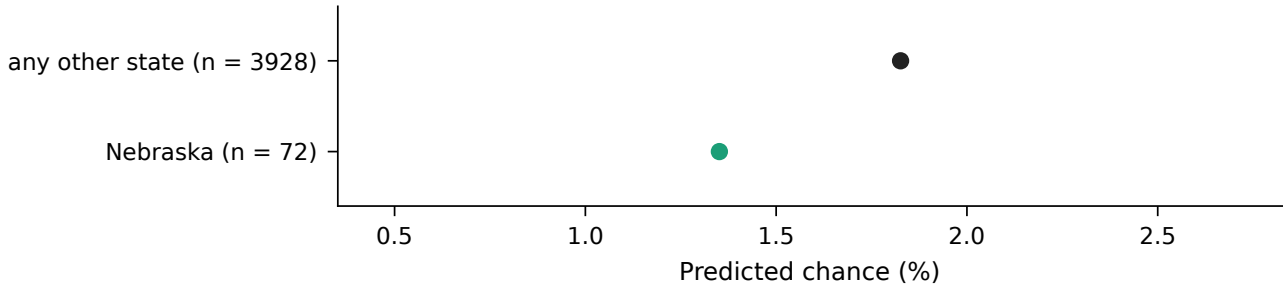


Figure 24: Early rise in calls: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

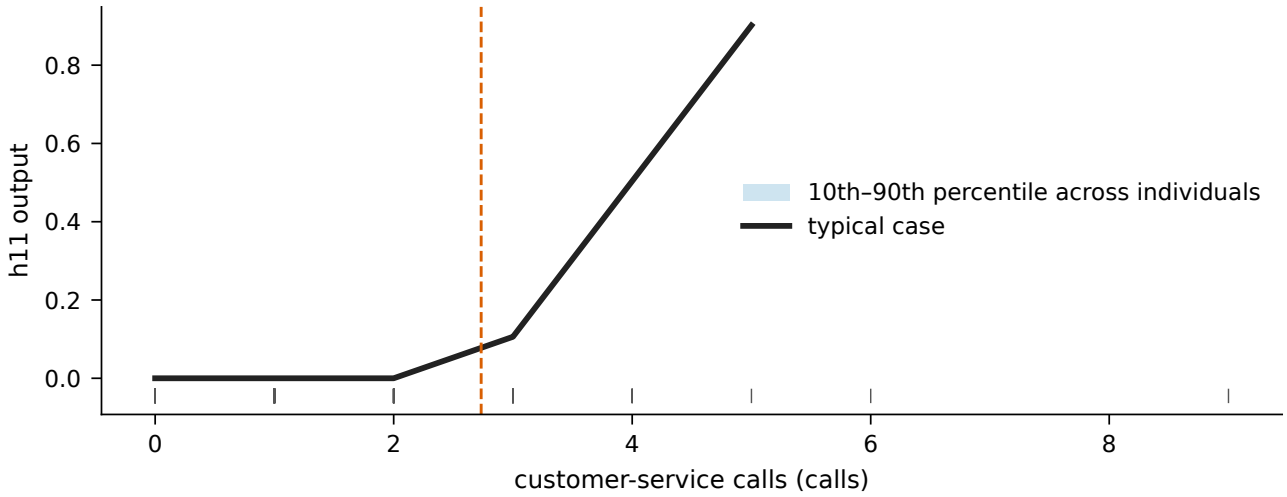


Figure 25: Early rise in calls: This part's own output, unit h11 (the output of 'Early rise in calls'), on its own scale (its activation), with every other input at the typical case; it is active for 830 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

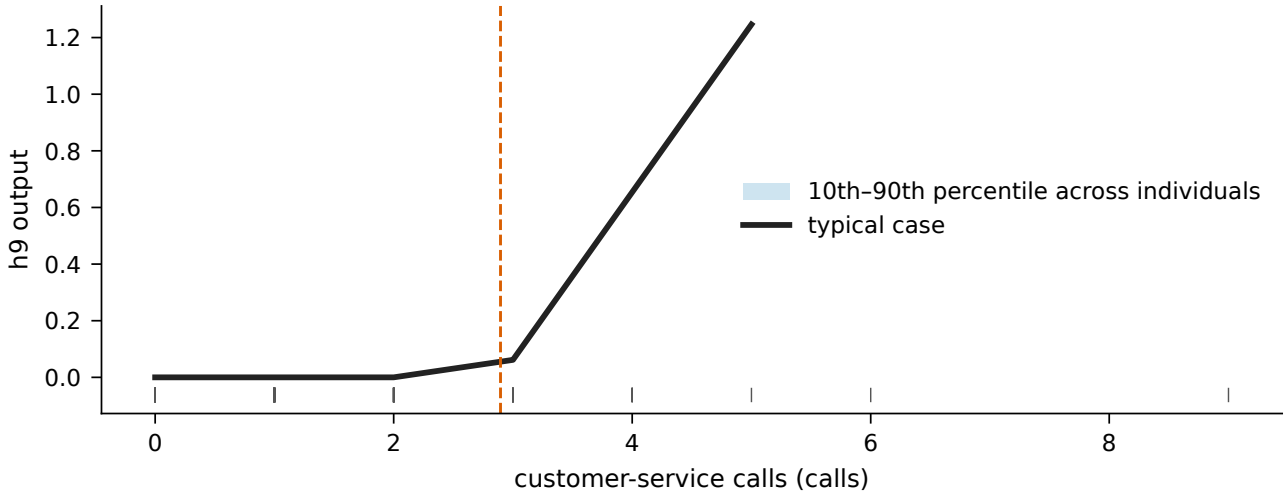


Figure 26: Early rise in calls: This part's own output, unit h9 (the output of 'Early rise in calls'), on its own scale (its activation), with every other input at the typical case; it is active for 830 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

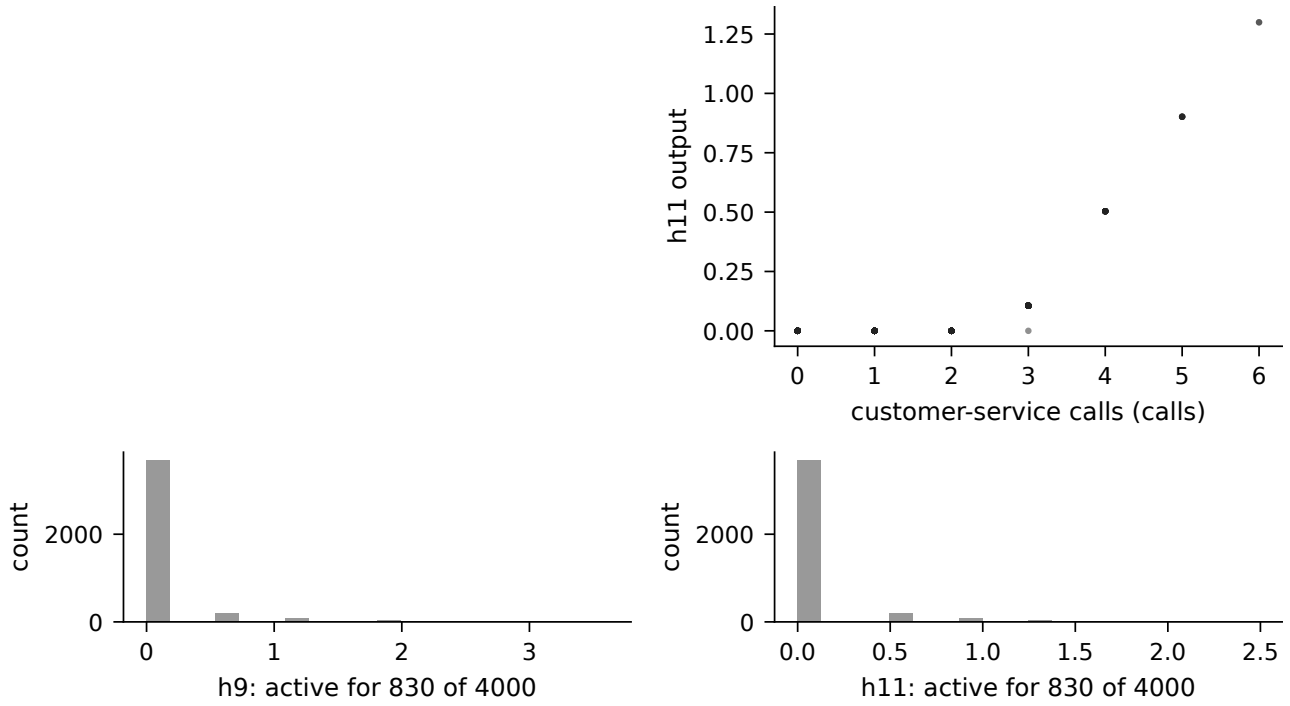


Figure 27: Early rise in calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.4.

1.2.5 Late rise in calls

Part of: Service calls (1.2.3).
Figure 28 shows what feeds this part; Figure 29 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

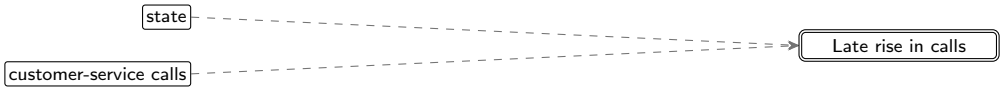


Figure 28: Late rise in calls: what feeds it, one level down.

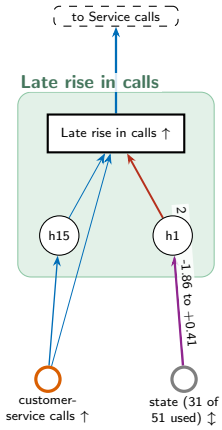


Figure 29: Late rise in calls: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving for a small group of customers, more in Colorado and less in California. [C122, C117, C118]

Evidence. It acts on 134 of 4000 training customers, 74 of whom left; for the typical customer it acts on, the chance is about 39% without it and about 75% with it. [C122] For the typical customer it acts on, the chance through it is about 75% in Colorado and about 39% in California, against about 52% in other states. [C117, C118] Its formula reproduces the part exactly on all 4000 training customers. [C123]
Evidence plots: Figure 30, Figure 31, Figure 32, Figure 33, Figure 34.
Figure 33 shows this part's own output, unit h6: it switches on at about 5 calls of customer-service calls, and is zero below.

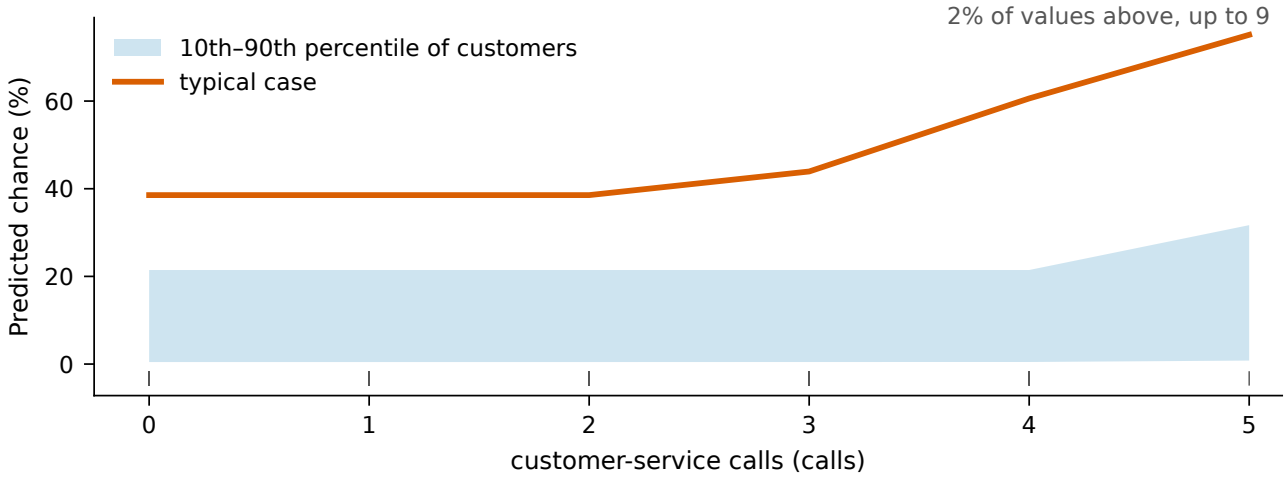


Figure 30: Late rise in calls: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

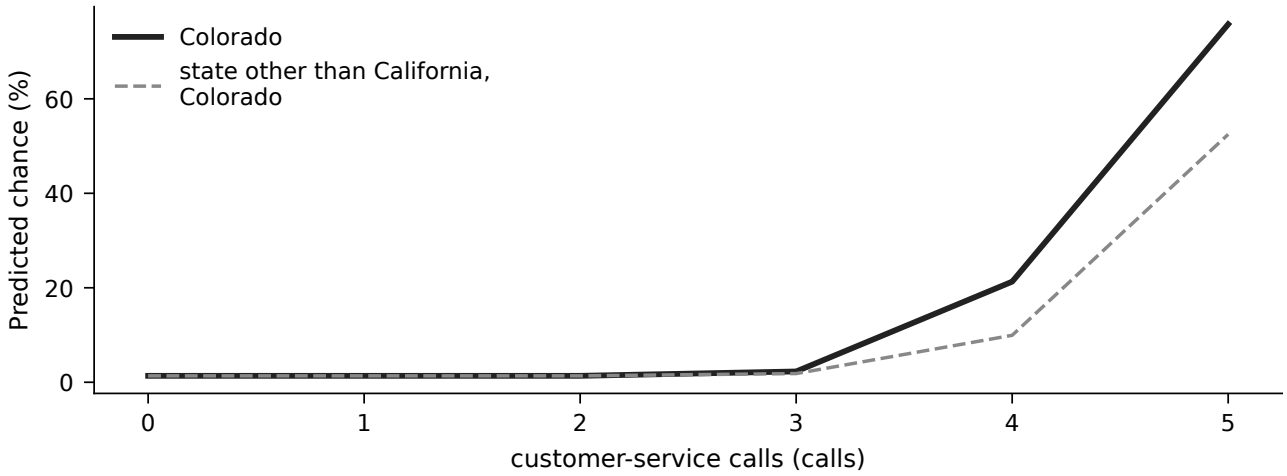


Figure 31: Late rise in calls: How the effect differs between groups: one curve per group, for the typical case of each group.

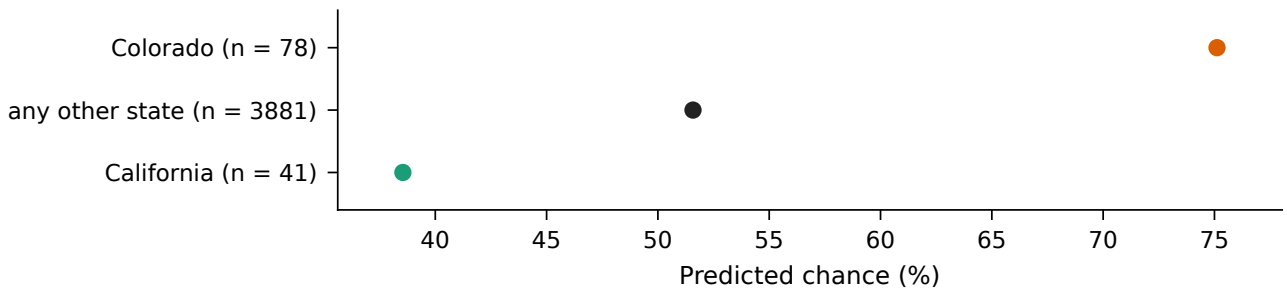


Figure 32: Late rise in calls: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

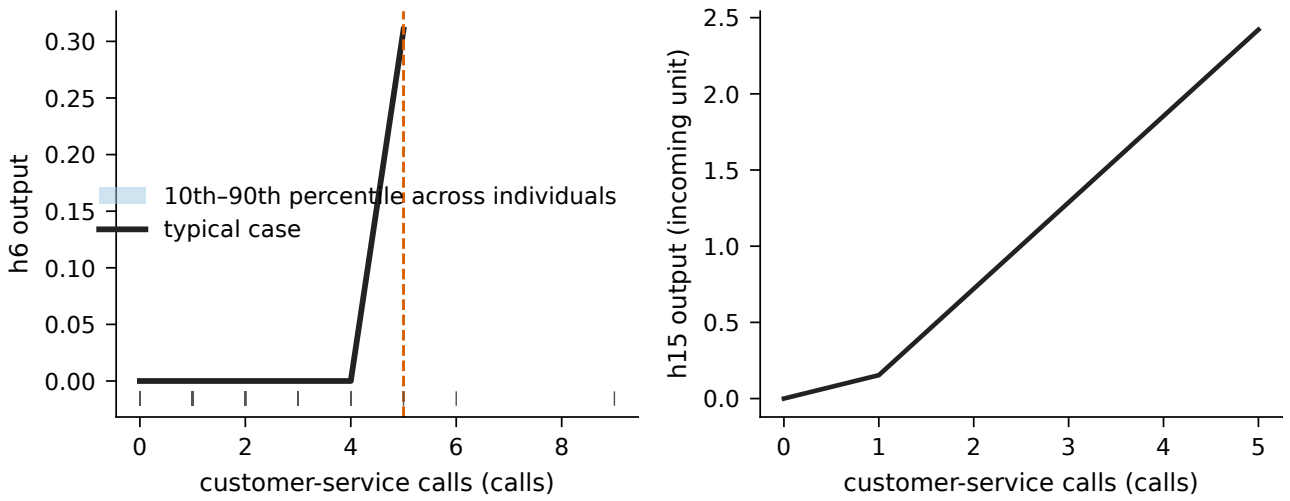


Figure 33: Late rise in calls: This part's own output, unit h6 (combines 2 non-linear units), on its own scale (its activation), with every other input at the typical case; it is active for 134 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

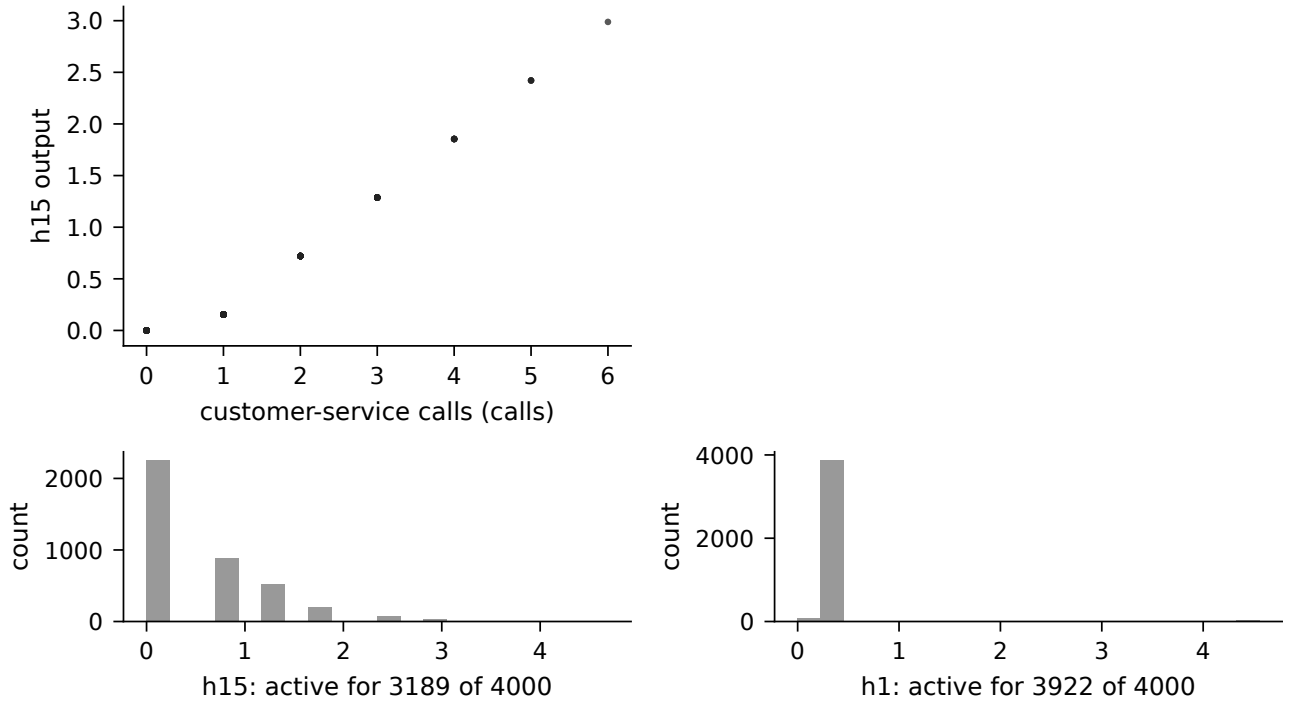


Figure 34: Late rise in calls: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.5.

1.2.6 Heavy daytime use

Part of: The whole model (1.2.1).
Figure 35 shows what feeds this part; Figure 36 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).

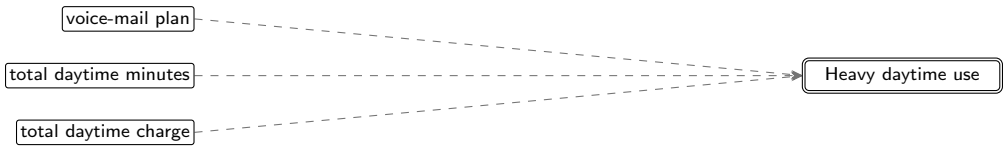


Figure 35: Heavy daytime use: what feeds it, one level down.

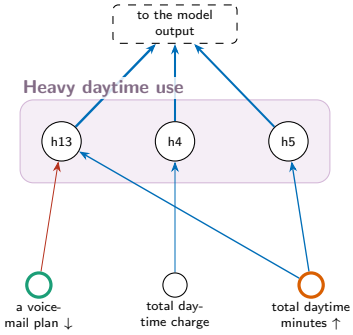


Figure 36: Heavy daytime use: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part raises the chance of leaving for heavy daytime users: daytime minutes add to it only above about 209 minutes, in an uneven way, and much less with a voice-mail plan. [C100, C099]

Evidence. This part acts on 1074 of 4000 training customers, 290 of whom left; for the typical customer it acts on, the chance is about 1% without it and about 4% with it. [C102] Daytime minutes add to the chance only once they rise above about 209 minutes, in an uneven way: for a typical such customer, from about 1% at 0 minutes to about 16% at 271 minutes, flattening out near the bottom because the predicted chance cannot go much lower. [C100] The rise is much larger without a voice-mail plan: from 112 to 249 daytime minutes the chance goes from about 1% to about 6% without one, but stays below 1% with one, in an uneven way, flattening out near the bottom because the predicted chance cannot go much lower. [C099] A voice-mail plan lowers it: for the typical customer it acts on, from about 4% to about 2%. [C098] Its formula reproduces the part exactly on all 4000 training customers. [C103]
Evidence plots: Figure 37, Figure 38, Figure 39, Figure 40, Figure 41, Figure 42.
Figure 40 shows this part's own output, unit h13: it switches on at about 209 minutes of total daytime minutes, and is zero below.
Figure 41 shows this part's own output, unit h5: it switches on at about 223.9 minutes of total daytime minutes, and is zero below.

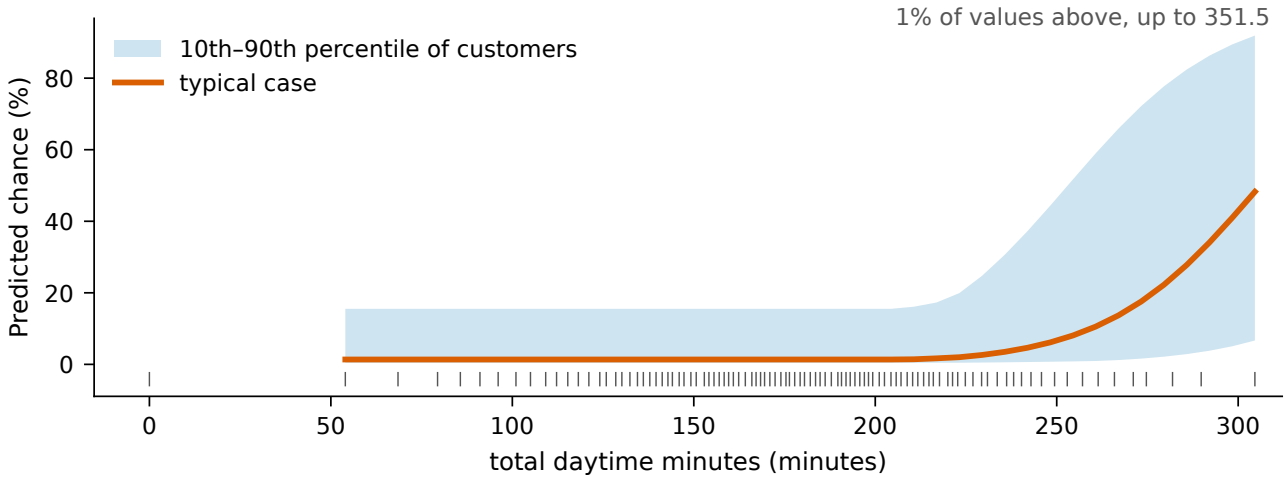


Figure 37: Heavy daytime use: Response curves through this part: the typical case (bold), the 10th to 90th percentile of individual customers (band), the fold or step point (dashed) with its spread across them (grey), and the observed values (ticks).

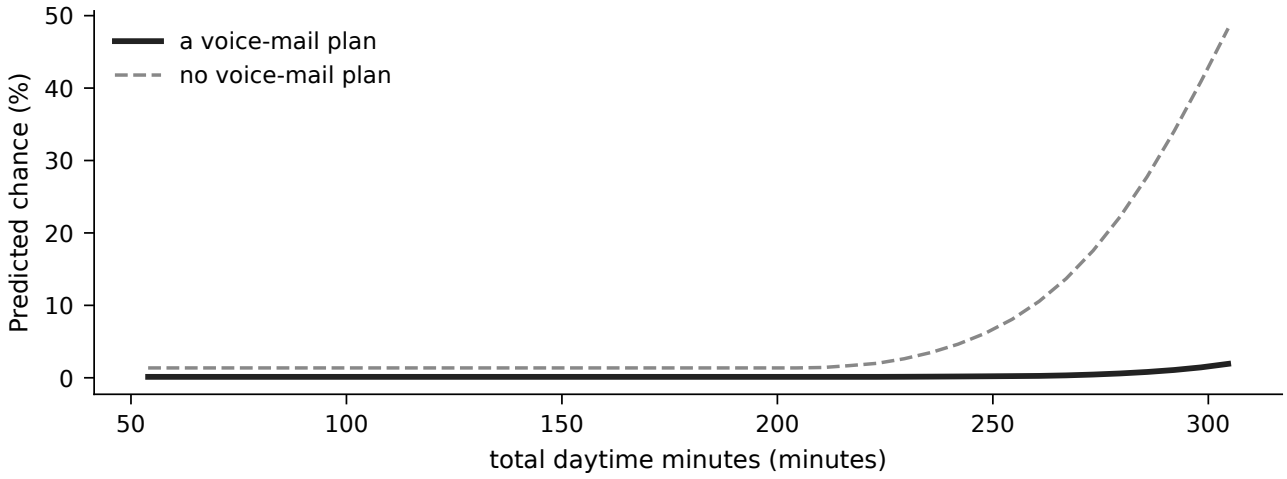


Figure 38: Heavy daytime use: How the effect differs between groups: one curve per group, for the typical case of each group.

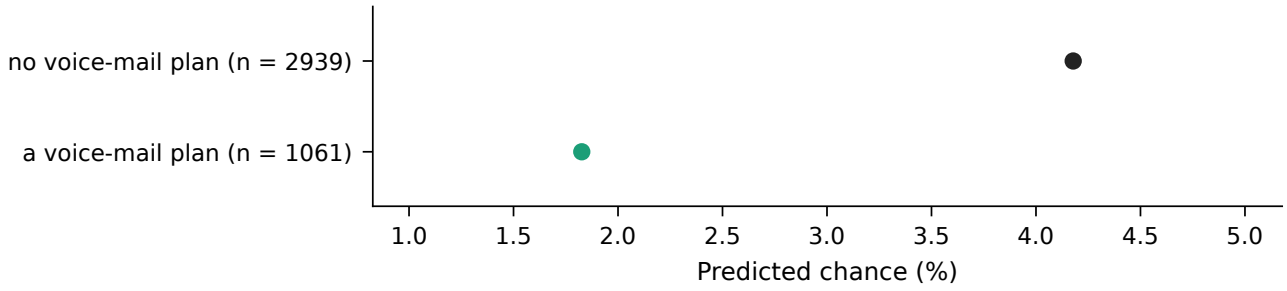


Figure 39: Heavy daytime use: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

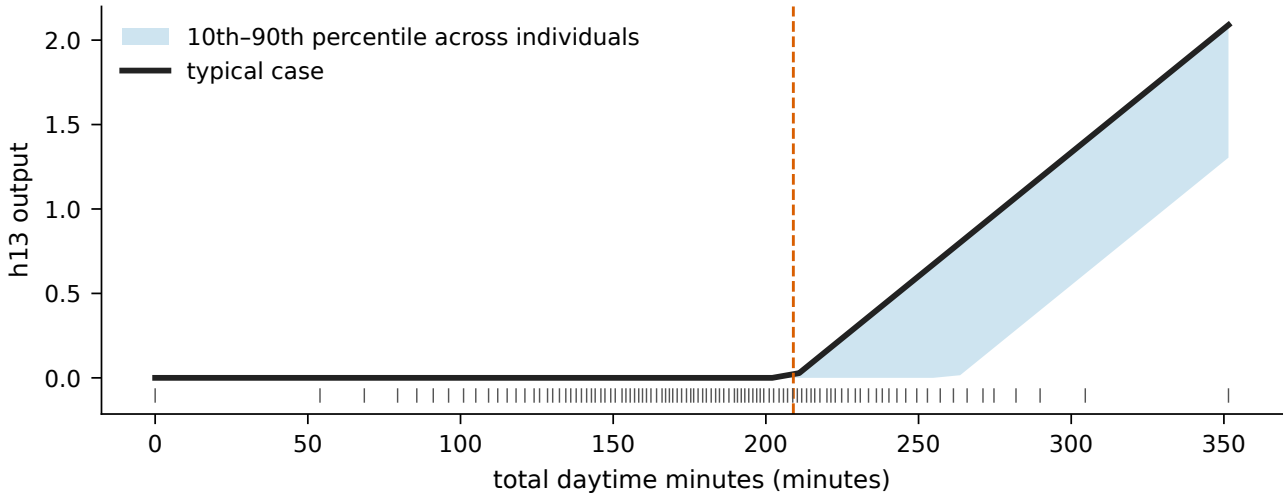


Figure 40: Heavy daytime use: This part's own output, unit h13 (the output of 'Heavy daytime use'), on its own scale (its activation), with every other input at the typical case; it is active for 929 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

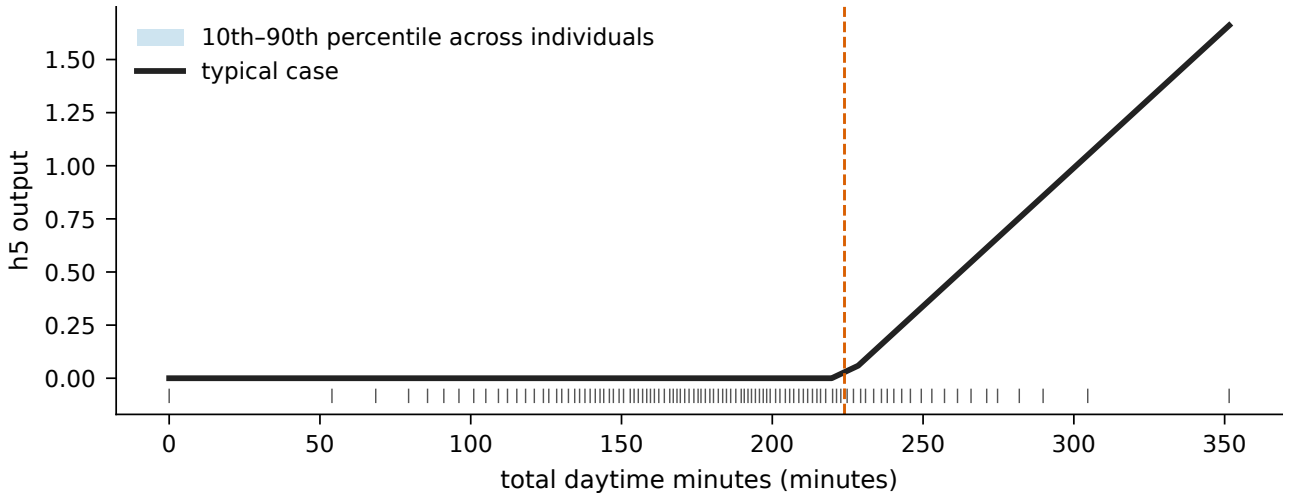


Figure 41: Heavy daytime use: This part's own output, unit h5 (the output of 'Heavy daytime use'), on its own scale (its activation), with every other input at the typical case; it is active for 818 of 4000. Bold: the typical case; band: 10th to 90th percentile across individuals; dashed: where it switches on; ticks: the data.

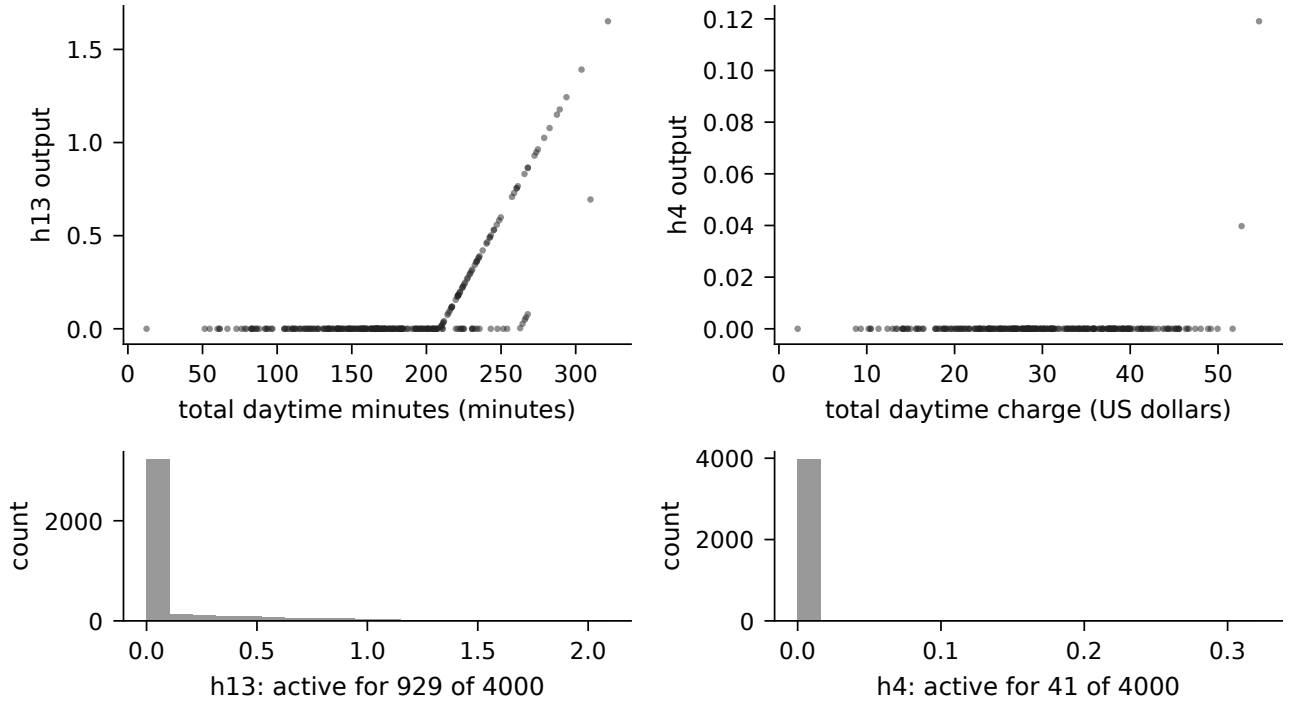


Figure 42: Heavy daytime use: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.6.

1.2.7 State units

Part of: The whole model (1.2.1).
Figure 43 shows what feeds this part; Figure 44 its units and every connection into them (dashed boxes: connections from or to other parts, drawn in their own sections).



Figure 43: State units: what feeds it, one level down.

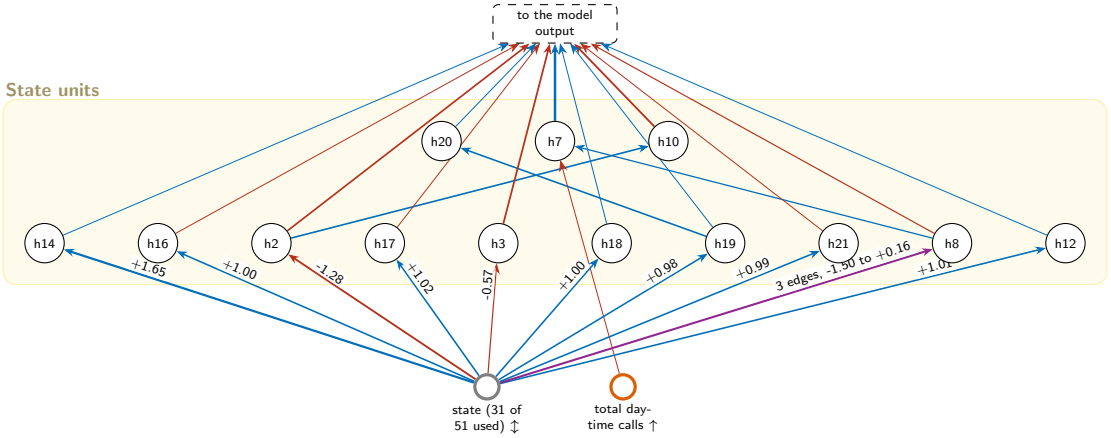


Figure 44: State units: its units and every connection into them. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends.

Hypothesis. This part reads only the state: it raises the chance of leaving in Rhode Island, Vermont and Tennessee, and lowers it in Virginia and Nevada. [C104, C105, C106, C107, C108]

Evidence. This part reads only the state: it raises a typical customer's chance from about 1% to about 12% in Rhode Island, to about 3% in Vermont and to about 2% in Tennessee. [C104, C105, C106] It lowers the chance to below 1% in Virginia and in Nevada. [C107, C108] Its formula reproduces the part exactly on all 4000 training customers. [C116]
Evidence plots: Figure 45, Figure 46.

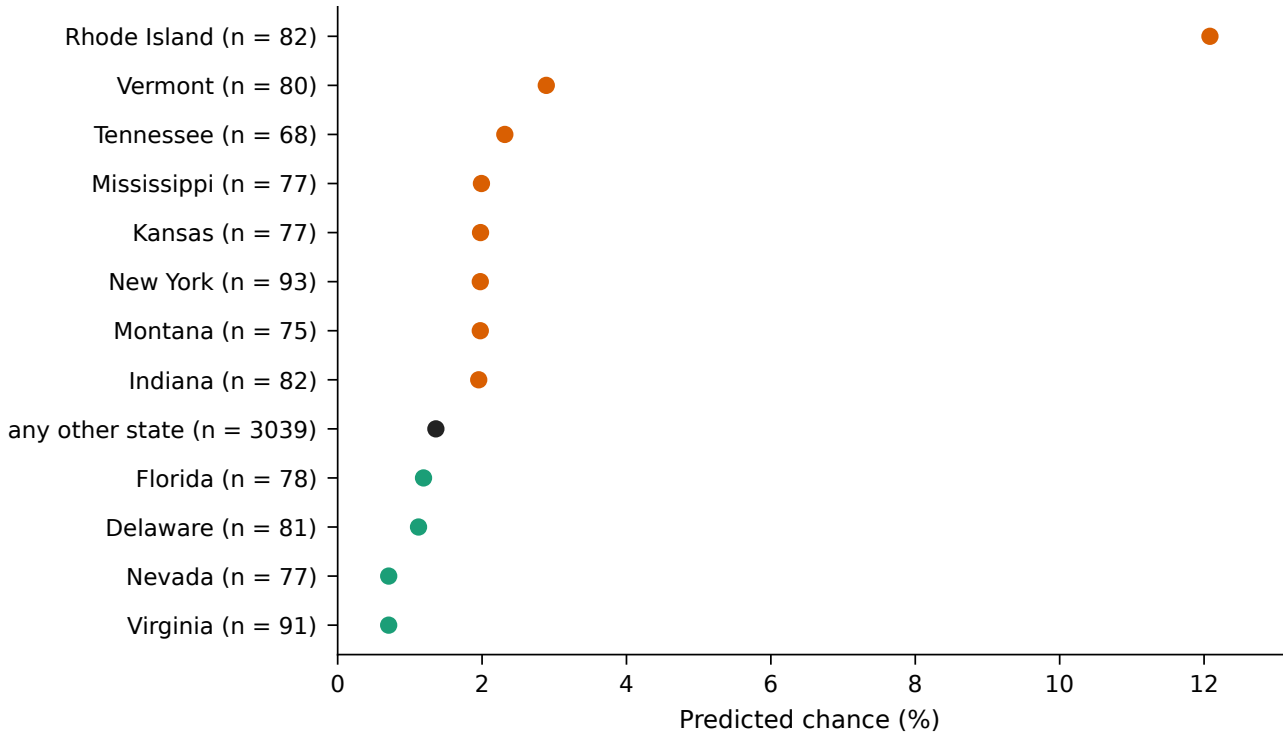


Figure 45: State units: The predicted value for the typical case at each level, through this part (n = training customers; grey: fewer than 20).

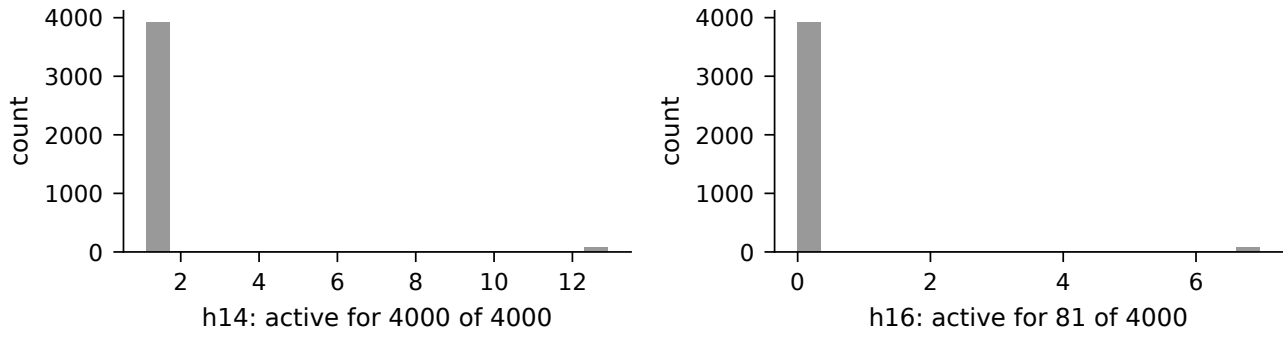


Figure 46: State units: This part's hidden units: each unit's output against its main numeric input (top) and how often it is active (bottom).

Technical detail: T.7.

1.3 Not supported by the model

- C101: threshold claim about total daytime charge in part ann_85: not supported by the model (the effect past the fold is too small or not consistent across patients).
- C121: threshold claim about number of calls to customer service in part ann_83: not supported by the model (the effect past the fold is too small or not consistent across patients).

Technical detail: churn100-s4-c3

The record behind each part: the original hypothesis, the closed form, and every claim with its estimate (log-odds and odds ratio), range across patients, n and test check.

T.1 The whole model

Annotation ann_88 (churn_logit). Reads: state, account length, area code, international calling plan, voice-mail plan, number of voice-mail messages, total daytime minutes, total daytime calls, total daytime charge, total evening minutes, total evening calls, total evening charge, total night-time minutes, total night-time calls, total night-time charge, total international minutes, total international calls, total international charge, number of calls to customer service.

Original hypothesis (unchanged). Root: logit P(churn) = -1.35 + linear_backbone (identity_lin) + service_calls (identity_svc) + 1.48*3118 + 1.50*4000 + 1.89*10131 (day_usage_hinges) + the state_offsets units (-1.11*1295 - 1.25*7872 - 1.17*1390 - 0.56*5786 + small terms; 5533 is never active), purely additive in the logit; the sigmoid is the only coupling. 30 inputs are unconnected (international minutes, calls and charge, night minutes/charge/calls, evening minutes and calls, area codes 408/510 and 20 state indicators) and have zero effect. Train AUC 0.8884, accuracy 0.8795; the mean predicted churn is 0.082 against a base rate of 0.1415 (the model ranks well but under-predicts). Performance sources (train): all children replaced by least-squares linear fits gives AUC 0.8352 (an L2 logistic regression refit on all 71 inputs gets 0.8433); restoring day_usage_hinges alone gives 0.8713, service_calls alone 0.8657, both 0.8884; the state units restore nothing (exactly linear). So the gain over linear (+0.053) comes from the day-usage and service-call thresholds, which are complementary. Rare-category switches: state:11 -2.50 logit (73 rows, empirical -1.35); state:46 -1.46 net (80 rows, -0.91); state:47 +1.35 (72 rows, +0.92); state:0 -0.97 (56, -0.76); state:14 -0.94 (68, -0.53); state:29 -0.53 direct plus switching off the early service-call hinge (mean -0.78, down to -6.15 for many-call customers; 72 rows, -0.79); state:39 -0.55 net (82, -0.93: direct -2.85 mostly cancelled by the state units); state:33 +0.38 net (direct +1.05, unit -0.68); state:18 +0.36 (65 rows, empirical -0.48: opposite sign); state:4 switches off the late chain (41 rows, empirical +0.80).

Closed form. Read through its parts (their formulas are in their own sections).

Split inputs. To annotate the parts, the explanation split these inputs into copies (a refactoring step that does not change what the model computes); the diagrams draw each input once: customer-service calls (2 copies); total daytime calls (2 copies); a voice-mail plan (2 copies).

State: every level the model reads. Whole-model change in the predicted chance for a typical customer when only the state changes from the reference; n = training customers in that level.

Claim	From
C004	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C006	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C010	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C011	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C017	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C018	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C019	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C020	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C021	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C022	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C023	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C024	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C026	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C032	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C035	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C036	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C039	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C040	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C037	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C034	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C033	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C028	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C027	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C025	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C016	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C015	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C014	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C013	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C012	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C009	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North
C007	state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 18%	2.74	15.6	2.74 to 2.74	3620, 380	agrees; AUC gain 0.066 [0.035, 0.098]	yes
C002	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 6%	1.61	5.00	1.61 to 1.61	4000	agrees; AUC gain 0.014 [0.0002, 0.029]	yes
C003	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 6%	1.30	3.67	0.49 to 1.59	4000	agrees; AUC gain 0.074 [0.044, 0.105]	yes
C004	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Washington (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 5%	1.35	3.86	1.35 to 1.35	1693, 72	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C005	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	1.01	2.73	1.01 to 1.01	4000	agrees; AUC gain 0.014 [0.003, 0.026]	yes
C006	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Maine (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 3%	0.71	2.04	0.71 to 0.71	1693, 86	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C007	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Hawaii (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to <1%	-2.50	0.08	-2.50 to -2.50	1693, 73	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C008	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	-2.62	0.07	-3.50 to -2.38	2939, 1061	agrees; AUC gain 0.049 [0.028, 0.072]	yes
C009	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Vermont (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to <1%	-1.46	0.23	-1.46 to -1.46	1693, 80	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C010	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Tennessee (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.54	1.72	0.54 to 0.54	1693, 68	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C011	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to South Carolina (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.54	1.71	0.54 to 0.54	1693, 66	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C012	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Alaska (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.97	0.38	-0.97 to -0.97	1693, 56	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C013	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Illinois (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.94	0.39	-0.94 to -0.94	1693, 68	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C014	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Pennsylvania (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.77	0.46	-0.77 to -0.77	1693, 56	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C015	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Virginia (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.66	0.52	-0.66 to -0.66	1693, 91	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C016	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Georgia (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.64	0.53	-0.64 to -0.64	1693, 60	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C017	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Mississippi (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.39	1.47	0.39 to 0.39	1693, 77	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C018	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Nevada (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.38	1.47	0.38 to 0.38	1693, 77	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C019	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Kansas (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.38	1.46	0.38 to 0.38	1693, 77	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C020	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Montana (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.38	1.46	0.38 to 0.38	1693, 75	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C021	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to New York (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.38	1.46	0.38 to 0.38	1693, 93	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C022	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Indiana (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.37	1.45	0.37 to 0.37	1693, 82	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C023	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Louisiana (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.36	1.44	0.36 to 0.36	1693, 65	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C024	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Utah (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.36	1.44	0.36 to 0.36	1693, 89	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C025	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Rhode Island (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.55	0.58	-0.55 to -0.55	1693, 82	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C026	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Massachusetts (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.35	1.42	0.35 to 0.35	1693, 86	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C027	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Nebraska (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.81	0.44	-0.83 to -0.53	1693, 72	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C028	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Florida (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.44	0.64	-0.44 to -0.44	1693, 78	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C029	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 99% of patients same; verified; risk saturates near 0%*	1% to 2%	0.30	1.36	0.31 to 0.31	4000	agrees; AUC gain 0.106 [0.065, 0.145]	yes
C030	direction: total daytime calls: 75 calls to 125 calls (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	0.32	1.37	0.32 to 0.32	4000	disagrees; AUC gain -0.002 [-0.005, 0.001]	yes
C031	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	0.28	1.32	0.28 to 0.28	4000	disagrees; AUC gain -0.001 [-0.004, 0.001]	yes
C032	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Oregon (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 2%	0.23	1.26	0.23 to 0.23	1693, 84	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C033	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Delaware (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.20	0.82	-0.20 to -0.20	1693, 81	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C034	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to North Carolina (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.13	0.87	-0.13 to -0.13	1693, 74	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C035	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to North Dakota (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	0.06	1.07	0.06 to 0.06	1693, 68	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C036	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Oklahoma (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	0.03	1.03	0.03 to 0.03	1693, 79	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C037	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Arizona (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.02	0.98	-0.02 to -0.02	1693, 73	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C038	direction: area code: area code 415 to area code other than area code 415 (whole model) *shape: categorical; highest to lowest: area code 415, area code other than area code 415*	1% to 1%	-0.02	0.98	-0.02 to -0.02	2016, 1984	disagrees; AUC gain -7e-05 [-0.0004, 0.0003]	yes
C039	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to California (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	-0.03	0.97	0.00 to 0.00	1693, 41	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C040	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Colorado (whole model) *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 1%	0.11	1.11	0.00 to 0.22	1693, 78	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C041	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 18%	2.74	15.6		3620, 380	agrees; AUC gain 0.066 [0.035, 0.098]	yes
C042	magnitude: voice-mail messages: 0 messages to 32 messages, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 6%	1.61	5.00		4000	agrees; AUC gain 0.014 [0.0002, 0.029]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C043	magnitude: total daytime minutes: 112 minutes to 249 minutes, typical patient *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 6%	1.59	4.92		4000	agrees; AUC gain 0.074 [0.044, 0.105]	yes
C044	magnitude: state: state other than Alaska, Georgia, Hawaii, Illinois, Indiana, Kansas, Louisiana, Massachusetts, Maine, Mississippi, Montana, North Carolina, North Dakota, Nebraska, Arizona, Nevada, New York, Oklahoma, Oregon, Pennsylvania, Rhode Island, California, South Carolina, Tennessee, Utah, Virginia, Vermont, Washington, Colorado, Delaware, Florida to Washington, typical patient *shape: categorical; highest to lowest: Washington, Maine, Tennessee, South Carolina, Mississippi, Nevada …*	1% to 5%	1.35	3.86		1693, 72	disagrees; AUC gain -0.013 [-0.028, -0.0007]	yes
C045	magnitude: total evening charge: 11.76 US dollars to 22.57 US dollars, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	1.01	2.73		4000	agrees; AUC gain 0.014 [0.003, 0.026]	yes
C046	null: model ignores total evening calls (exact rule)		0.00			4000	train only	yes
C047	null: model ignores total evening minutes (exact rule)		0.00			4000	train only	yes
C048	null: model ignores total international calls (exact rule)		0.00			4000	train only	yes
C049	null: model ignores total international charge (exact rule)		0.00			4000	train only	yes
C050	null: model ignores total international minutes (exact rule)		0.00			4000	train only	yes
C051	null: model ignores total night-time calls (exact rule)		0.00			4000	train only	yes
C052	null: model ignores total night-time charge (exact rule)		0.00			4000	train only	yes
C053	null: model ignores total night-time minutes (exact rule)		0.00			4000	train only	yes
C054	structure: made of linear_backbone, service_calls, day_usage_hinges, state_offsets; added					4000	train only	yes

T.2 Fixed-weight inputs

Annotation ann_82 (linear_backbone). Reads: state, account length, area code, international calling plan, voice-mail plan, number of voice-mail messages, total daytime calls, total evening charge.

Original hypothesis (unchanged). Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 28 inputs: voicemail plan -1.05 per SD (-2.61 logit for having a plan), international plan +0.80 (+2.75 logit for having one), voicemail messages +0.68 per SD (offsets most of the plan effect for heavy message users), evening charge +0.40, day calls +0.13, account length +0.11, and state offsets: state:39 -0.40 per SD (-2.85 logit, but state_offsets gives back +2.30), state:11 -0.33 (-2.50), state:46 -0.31 (-2.22, of which state_offsets gives back +0.77), state:47 +0.18 (+1.35), state:33 +0.14 (+1.05, of which state_offsets takes back -0.68), state:0 -0.97, state:14 -0.94, state:38 -0.77, state:21 +0.71, state:10 -0.64, state:40 +0.54, state:29 -0.53, then smaller states and area code 415 (+0.01). Day minutes, day charge and customer-service calls have NO direct edge: they act only through hinges. Evidence: closed form exact (identity); train sensitivities equal the weights (voicemail plan 1.05, international plan 0.80, voicemail messages 0.68, state:39 0.40, evening charge 0.40).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.77636e-15).

$$
\begin{aligned}
y_0 = {} & 0.0337x_{state:37} + 0.144x_{state:33_a} + 0.679x_{numberVmailMessages} \\
& + 0.0685x_{state:40} + 0.00461x_{state:36} + 0.18x_{state:47} + 0.398x_{totalEveCharge} \\
& + 0.103x_{state:21} + 0.126x_{totalDayCalls_a} + 0.00838x_{state:28} \\
& + 0.00889x_{areaCode:415} + 0.0459x_{state:18} + 0.805x_{internationalPlan} \\
& + 0.0513x_{state:19} + 0.106x_{accountLength} + 0.0534x_{state:44} \\
& - 1.05x_{voiceMailPlan_a} - 0.335x_{state:11} - 0.404x_{state:39_a} \\
& - 0.311x_{state:46_a} - 0.0781x_{state:10} - 0.0904x_{state:38} - 0.00327x_{state:3} \\
& - 0.114x_{state:0} - 0.07x_{state:29_a} - 0.0181x_{state:27} - 0.0419x_{state:9_a} \\
& - 0.122x_{state:14}
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C055	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 18%	2.74	15.6	2.74 to 2.74	3620, 380	agrees; AUC gain 0.066 [0.035, 0.098]	yes
C056	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 6%	1.61	5.00	1.61 to 1.61	4000	agrees; AUC gain 0.014 [0.0002, 0.029]	yes
C057	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Washington (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 5%	1.35	3.86	1.35 to 1.35	2456, 72	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C058	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Nevada (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 4%	1.05	2.85	1.05 to 1.05	2456, 77	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C059	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	1.01	2.73	1.01 to 1.01	4000	agrees; AUC gain 0.014 [0.003, 0.026]	yes
C060	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Maine (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 3%	0.71	2.04	0.71 to 0.71	2456, 86	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C061	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Rhode Island (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to <1%	-2.85	0.06	-2.85 to -2.85	2456, 82	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C062	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Hawaii (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to <1%	-2.50	0.08	-2.50 to -2.50	2456, 73	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C063	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	-2.38	0.09	-2.38 to -2.38	2939, 1061	agrees; AUC gain 0.038 [0.017, 0.061]	yes
C064	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Vermont (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to <1%	-2.22	0.11	-2.22 to -2.22	2456, 80	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C065	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to South Carolina (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 2%	0.54	1.71	0.54 to 0.54	2456, 66	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C066	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Alaska (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.97	0.38	-0.97 to -0.97	2456, 56	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C067	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Illinois (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.94	0.39	-0.94 to -0.94	2456, 68	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C068	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Pennsylvania (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.77	0.46	-0.77 to -0.77	2456, 56	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C069	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Georgia (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.64	0.53	-0.64 to -0.64	2456, 60	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C070	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Louisiana (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 2%	0.36	1.44	0.36 to 0.36	2456, 65	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C071	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Utah (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 2%	0.36	1.44	0.36 to 0.36	2456, 89	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C072	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Massachusetts (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 2%	0.35	1.42	0.35 to 0.35	2456, 86	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C073	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Nebraska (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.53	0.59	-0.53 to -0.53	2456, 72	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C074	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	0.32	1.37	0.32 to 0.32	4000	disagrees; AUC gain -0.002 [-0.005, 0.001]	yes
C075	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	0.28	1.32	0.28 to 0.28	4000	disagrees; AUC gain -0.001 [-0.004, 0.001]	yes
C076	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Oregon (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 2%	0.23	1.26	0.23 to 0.23	2456, 84	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C077	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Florida (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.30	0.74	-0.30 to -0.30	2456, 78	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C078	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to North Carolina (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.13	0.87	-0.13 to -0.13	2456, 74	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C079	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to North Dakota (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	0.06	1.07	0.06 to 0.06	2456, 68	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C080	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Oklahoma (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	0.03	1.03	0.03 to 0.03	2456, 79	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C081	direction: state: state other than Alaska, Georgia, Hawaii, Illinois, Louisiana, Massachusetts, Maine, North Carolina, North Dakota, Nebraska, Arizona, Nevada, Oklahoma, Oregon, Pennsylvania, Rhode Island, South Carolina, Utah, Vermont, Washington, Florida to Arizona (through this part) *shape: categorical; highest to lowest: Washington, Nevada, Maine, South Carolina, Louisiana, Utah …*	1% to 1%	-0.02	0.98	-0.02 to -0.02	2456, 73	disagrees; AUC gain -0.013 [-0.030, 0.002]	yes
C082	direction: area code: area code 415 to area code other than area code 415 (through this part) *shape: categorical; highest to lowest: area code 415, area code other than area code 415*	1% to 1%	-0.02	0.98	-0.02 to -0.02	2016, 1984	disagrees; AUC gain -7e-05 [-0.0004, 0.0003]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C083	structure: its formula reproduces it on 4000 training patients (largest difference 1.77636e-15)		2e-15			4000	train only	yes

T.3 Service calls

Annotation ann_87 (service_calls). Reads: state, number of calls to customer service.

Original hypothesis (unchanged). Composition of csc_early_hinge and csc_late_chain: their exits are summed by identity_svc = 1.88*9436 + 1.73*6713 + 1.71*5268, which enters the logit with weight 1; they read the same input (two copies of customer-service calls) plus state:29 (early hinge) and state:4/state:5 (late chain). Hypothesis: all of the model's service-call effect is a threshold: nothing for 0-2 calls, +0.30 logit at 3 calls, then about +2.3 logit per extra call (+2.06 at 4, +4.40 at 5, +6.67 at 6, +9.28 at 7); the early hinge carries most of it, the late chain adds an extra bend from 5 calls. For state:29 customers only the late chain remains. Evidence (train): churn by calls 0-7 is 0.115/0.110/0.110/0.106/0.437/0.571/0.679/0.545 against model 0.058/0.055/0.050/0.062/0.209/0.573/0.890/0.979: the threshold is right, but the model under-predicts up to 4 calls and is over-confident from 6 calls (42 customers). Sensitivity of identity_svc: state:29 1.04, calls (early copy) 0.48, state:4 0.13. Mean-ablating both children gives AUC 0.8016, linear replacement 0.8713; restoring service_calls alone onto the all-linear model gives 0.8657 (all-linear 0.8352).

Closed form. Read through its parts (their formulas are in their own sections).

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C084	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 99% of patients same; verified; risk saturates near 0%*	1% to 2%	0.30	1.36	0.31 to 0.31	4000	agrees; AUC gain 0.106 [0.065, 0.145]	yes
C085	direction: state: state other than Nebraska, California, Colorado to Nebraska (through this part) *shape: categorical; highest to lowest: Nebraska, California, Colorado, state other than Nebraska, California, Colorado*	1% to 1%	-0.29	0.75	-0.31 to 0.00	3809, 72	unclear; AUC gain 0.0001 [-0.0007, 0.002]	yes
C086	direction: state: state other than Nebraska, California, Colorado to California (through this part) *shape: categorical; highest to lowest: Nebraska, California, Colorado, state other than Nebraska, California, Colorado*	1% to 1%	-0.03	0.97	0.00 to 0.00	3809, 41	unclear; AUC gain 0.0001 [-0.0007, 0.002]	yes
C087	direction: state: state other than Nebraska, California, Colorado to Colorado (through this part) *shape: categorical; highest to lowest: Nebraska, California, Colorado, state other than Nebraska, California, Colorado*	1% to 1%	0.11	1.11	0.00 to 0.22	3809, 78	unclear; AUC gain 0.0001 [-0.0007, 0.002]	yes
C088	interaction: effect of customer-service calls (0 calls to 3 calls) by state: Nebraska minus state other than Nebraska, California, Colorado; stronger in state other than Nebraska, California, Colorado *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	Nebraska: 1% to 1%; state other than Nebraska, California, Colorado: 1% to 2%	-0.31	0.74		72, 3809	agrees; AUC gain 0.0001 [-0.0007, 0.002]	yes
C089	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 99% of patients same; verified; risk saturates near 0%*	1% to 10% (typical patient with a fold)	2.06	7.84	2.08 to 2.08	4000, 831	agrees; AUC gain 0.106 [0.065, 0.145]	yes
C090	structure: made of csc_late_chain, csc_early_hinge; added					4000	train only	yes

T.4 Early rise in calls

Annotation ann_84 (csc_early_hinge). Reads: state, number of calls to customer service.

Original hypothesis (unchanged). 9436 = relu(-0.62 + 0.52 csc - 1.12 state:29), 6713 = relu(-0.10 + 1.49*9436); logit weights +1.88 and +1.73 (via identity_svc). Hypothesis: the main hinge on customer-service calls: off for 0-2 calls, just on at 3 calls (+0.30 logit), then about +1.8 logit per extra call (+2.04 at 4, +3.85 at 5, +5.43 at 6, +7.40 at 7); never on for the 72 state:29 customers. Active on 20.7% of customers (all with 3+ calls, outside state:29); empirical churn jumps from 0.11 at 3 calls to 0.44 at 4. Evidence: sensitivity 0.35 (6713) / 0.23 (9436) to state:29 and 0.16 / 0.11 to calls; the line_plot sweeps state:29 (first entry), so it is flat at 0 (tool quirk: line plots vary only the first entry). Mean ablation AUC 0.8281; linear replacement 0.8736 (R2 0.57/0.50); restored alone onto all-linear 0.8654.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 8.88178e-16).

$$
\begin{aligned}
y_0 &= \operatorname{relu}\big(-0.615 + 0.52 x_{numberCustomerServiceCalls_b} \\
&\qquad - 1.12 x_{state:29_b}\big) \\
y_1 &= \operatorname{relu}\big(-0.0961 + 1.49 \operatorname{relu}\big(-0.615 \\
&\qquad + 0.52 x_{numberCustomerServiceCalls_b} - 1.12 x_{state:29_b}\big)\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C091	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 2%	0.30	1.35	0.31 to 0.31	4000	agrees; AUC gain 0.082 [0.050, 0.114]	yes
C092	direction: state: state other than Nebraska to Nebraska (through this part) *shape: categorical; highest to lowest: state other than Nebraska, Nebraska*	2% to 1% (typical patient this part acts on)	-0.29	0.75	-0.31 to 0.00	3928, 72	unclear; AUC gain 8e-05 [-0.0002, 0.0003]	yes
C093	interaction: effect of customer-service calls (0 calls to 3 calls) by state: Nebraska minus state other than Nebraska; stronger in state other than Nebraska *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	Nebraska: 1% to 1%; state other than Nebraska: 1% to 2%	-0.31	0.73		72, 3928	agrees; AUC gain 8e-05 [-0.0002, 0.0003]	yes
C094	threshold: customer-service calls: fold at about 3 calls (starts) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 10% (typical patient with a fold)	2.08	8.00	2.08 to 2.08	3928, 830	agrees; AUC gain 0.082 [0.050, 0.114]	yes
C095	subgroup: who this part acts on: 830 of 4000 patients	1% to 2% (typical patient this part acts on (risk without the part, then with it))	1.36	3.88	0.31 to 3.85	830, 3170	agrees; AUC gain 0.083 [0.051, 0.116]	yes
C096	structure: its formula reproduces it on 4000 training patients (largest difference 8.88178e-16)		9e-16			4000	train only	yes

T.5 Late rise in calls

Annotation ann_83 (csc_late_chain). Reads: state, number of calls to customer service.

Original hypothesis (unchanged). 12374 = relu(0.48 + 0.74 csc) (always active except at 0 calls, where it is just below its kink), 349 = relu(0.22 + 0.41 state:4 - 1.86 state:5), 5268 = relu(-0.56 + 0.33 csc + 0.26*12374 - 1.39*349); logit weight +1.71 (via identity_svc). 349 takes three values: 0.44 for most customers, 0 in state:5 (78 customers) and 4.55 in state:4 (41 customers). Hypothesis: a late hinge on customer-service calls that is off for 0-4 calls for most customers and adds about +0.65 logit per call from 5 calls (+0.55 at 5, +1.24 at 6, +1.88 at 7); in state:5 it starts one call earlier (on from 3 calls) and in state:4 it is always off. Evidence (train, n=4000): 5268 is active on 3.4% of customers (on for 3% at 4 calls, 99% at 5); sensitivity of 5268 is 0.074 to state:4, 0.017 to calls, 0.017 to state:5; line plot correlation 0.51. Mean ablation AUC 0.8889 (full 0.8884): it adds nothing to ranking on its own; linear replacement 0.8887 (R2 0.27); restored alone onto the all-linearised model 0.8401 (all-linear 0.8352).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 2.22045e-16).

$$\begin{aligned} y_0 = \mathrm{relu}\big(&-0.557 + 0.256\,\mathrm{relu}\big(0.476 \\ &+ 0.74 x_{numberCustomerServiceCalls_ac}\big) + 0.327 x_{numberCustomerServiceCalls_ac} \\ &- 1.39\,\mathrm{relu}\big(0.219 + 0.414 x_{state:4} - 1.86 x_{state:5}\big)\big) \end{aligned}$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C117	direction: state: state other than California, Colorado to Colorado (through this part) *shape: categorical; highest to lowest: Colorado, state other than California, Colorado, California*	52% to 75% (typical patient this part acts on)	0.11	1.11	0.00 to 0.22	3881, 78	unclear; AUC gain 2e-05 [-0.0009, 0.001]	yes
C118	direction: state: state other than California, Colorado to California (through this part) *shape: categorical; highest to lowest: Colorado, state other than California, Colorado, California*	52% to 39% (typical patient this part acts on)	-0.03	0.97	0.00 to 0.00	3881, 41	unclear; AUC gain 2e-05 [-0.0009, 0.001]	yes
C119	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 2% of patients same; fallback*	39% to 44% (typical patient this part acts on)	0.004	1.00	0.00 to 0.00	4000	unclear; AUC gain 0.0003 [-0.0007, 0.002]	yes
C120	interaction: effect of customer-service calls (0 calls to 3 calls) by state: Colorado minus state other than California, Colorado; stronger in Colorado *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	Colorado: 1% to 2%; state other than California, Colorado: 1% to 2%	0.23	1.26		78, 3881	agrees; AUC gain 2e-05 [-0.0009, 0.001]	yes
C121	threshold: customer-service calls: fold at about 5 calls (starts) *shape: hinge at 4 calls; 98% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	0.02	1.02	0.00 to 0.00	3959, 134	unclear; AUC gain 0.0003 [-0.0007, 0.002]	no
C122	subgroup: who this part acts on: 134 of 4000 patients	39% to 75% (typical patient this part acts on (risk without the part, then with it))	0.85	2.35	0.53 to 1.88	134, 3866	unclear; AUC gain 0.0003 [-0.0008, 0.002]	yes
C123	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		2e-16			4000	train only	yes

T.6 Heavy daytime use

Annotation ann_85 (day_usage_hinges). Reads: voice-mail plan, total daytime minutes, total daytime charge.

Original hypothesis (unchanged). 10131 = relu(-0.63 + 0.79 dayMin - 0.35 voiceMailPlan), logit weight +1.89; 4000 = relu(-0.56 + 0.70 dayMin), weight +1.50; 3118 = relu(-0.84 + 0.36 dayCharge), weight +1.48. Day minutes and day charge are collinear (r = 1.0000), so all three are hinges in day usage. Hypothesis: day usage has no effect below about 200 minutes, then raises churn risk steeply in three stages: 10131 switches on at about 209 day minutes without a voicemail plan (about 263 with one), 4000 at about 224 minutes, and 3118 at a day charge of about 51.7 (about 304 minutes; on for only 1% of customers). Evidence (train): combined logit contribution 0 below 200 minutes, +0.05 at 200-220, +0.51 at 220-240, +1.31 at 240-260, +2.19 at 260-280, +3.10 at 280-300, +4.69 above 300 (voicemail-plan holders about 1.1-1.7 lower from 240 minutes up); churn by band 0.06/0.14/0.26/0.46/0.58/0.79 against model 0.04/0.08/0.15/0.24/0.37/0.64 (right shape, under-predicted, as is the whole model). Sensitivity: 10131 day minutes 0.18, voicemail plan 0.08; 4000 day minutes 0.14; 3118 day charge 0.004. Mean ablation AUC 0.8083 (the largest single drop); linear replacement 0.8657 (R2 0.47/0.44/0.05); restored alone onto all-linear 0.8713 (all-linear 0.8352).

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 2.22045e-16).

$$
\begin{aligned}
y_0 &= \mathrm{relu}\big(-0.836 + 0.365 x_{totalDayCharge}\big) \\
y_1 &= \mathrm{relu}\big(-0.565 + 0.7 x_{totalDayMinutes}\big) \\
y_2 &= \mathrm{relu}\big(-0.625 + 0.789 x_{totalDayMinutes} \\
&\qquad - 0.347 x_{voiceMailPlan_b}\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C097	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 6%	1.30	3.67	0.49 to 1.59	4000	agrees; AUC gain 0.074 [0.044, 0.105]	yes
C098	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	4% to 2% (typical patient this part acts on)	-0.24	0.79	-1.12 to 0.00	2939, 1061	agrees; AUC gain 0.007 [0.003, 0.011]	yes
C099	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by voice-mail plan: a voice-mail plan minus no voice-mail plan; stronger in no voice-mail plan *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	a voice-mail plan: <1% to <1%; no voice-mail plan: 1% to 6%	-1.11	0.33		1061, 2939	agrees; AUC gain 0.007 [0.003, 0.011]	yes
C100	threshold: total daytime minutes: fold at about 209 minutes (starts) *shape: accelerating; 100% of patients same; fallback; risk saturates near 0%*	1% to 16% (typical patient with a fold)	2.24	9.41	1.15 to 2.63	4000, 1074	agrees; AUC gain 0.074 [0.044, 0.105]	yes
C101	threshold: total daytime charge: fold at about 51.68 US dollars (starts) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	0.00	1.00	0.00 to 0.00	4000, 41	unclear; AUC gain 3e-05 [-3e-05, 0.0001]	no
C102	subgroup: who this part acts on: 1074 of 4000 patients	1% to 4% (typical patient this part acts on (risk without the part, then with it))	1.28	3.61	0.12 to 3.02	1074, 2926	agrees; AUC gain 0.080 [0.049, 0.113]	yes
C103	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		2e-16			4000	train only	yes

T.7 State units

Annotation ann_86 (state_offsets). Reads: state, total daytime calls.

Original hypothesis (unchanged). Thirteen units, each reading only one-hot state indicators, so each is exactly an offset per state (least-squares R2 = 1.0): 1295 -> 7872 (state:39; weights -1.11, -1.25: a constant -2.32 logit for everyone except state:39, where it is switched off, +2.30, against state:39's direct -2.85; net -0.55 vs empirical log-odds offset -0.93, 82 customers), 11361 (state:46, +0.77 against direct -2.22), 5786 (state:26/33/34: +0.38 for state:26 and state:34, -0.68 for state:33 against its direct +1.05), 1390 (state:25 +0.39), 17102 (state:45 -0.66), 15490 -> 15904 (state:42 +0.54), 15362 (state:16 +0.38), 9954 (state:15 +0.37), 12690 (state:8 -0.20), 13068 (state:9 -0.14, plus direct -0.30), and 5533, which also reads day calls but is never active on train (pre-activation at most -0.30), so it contributes exactly 0. Evidence: sensitivities are on the matching state only (state:39 1.25/1.28, state:46 1.65, state:42 0.98); linear replacement AUC 0.8884 (unchanged); mean ablation 0.8856.

Closed form (weights and biases at 3 significant figures; largest difference from the part on the training patients 1.11022e-16).

$$
\begin{aligned}
y_{0} &= \mathrm{relu}\big(0.78 - 1.28 x_{state:39_b}\big) \\
y_{1} &= \mathrm{relu}\big(0.0571 + 1.02\,\mathrm{relu}\big(0.78 \\
&\qquad - 1.28 x_{state:39_b}\big)\big) \\
y_{2} &= \mathrm{relu}\big(0.252 - 0.574 x_{state:25}\big) \\
y_{3} &= \mathrm{relu}\big(-0.38 + 1.01 x_{state:15}\big) \\
y_{4} &= \mathrm{relu}\big(1.36 + 1.65 x_{state:46_b}\big) \\
y_{5} &= \mathrm{relu}\big(-0.0337 + 1.0 x_{state:8}\big) \\
y_{6} &= \mathrm{relu}\big(-0.372 + 1.02 x_{state:9_b}\big) \\
y_{7} &= \mathrm{relu}\big(-0.581 + 0.999 x_{state:16}\big) \\
y_{8} &= \mathrm{relu}\big(0.361 + 0.984 x_{state:42}\big) \\
y_{9} &= \mathrm{relu}\big(-0.0356 + 0.993\,\mathrm{relu}\big(0.361 \\
&\qquad + 0.984 x_{state:42}\big)\big) \\
y_{10} &= \mathrm{relu}\big(-1.42 + 0.995 x_{state:45}\big) \\
y_{11} &= \mathrm{relu}\big(0.383 + 0.163 x_{state:33_b} - 1.5 x_{state:26} \\
&\qquad - 0.732 x_{state:34}\big) \\
y_{12} &= \mathrm{relu}\big(-1.54 + 0.655\,\mathrm{relu}\big(0.383 \\
&\qquad + 0.163 x_{state:33_b} - 1.5 x_{state:26} - 0.732 x_{state:34}\big) \\
&\qquad - 0.00543 x_{totalDayCalls_b}\big)
\end{aligned}
$$

Evidence table (next page, landscape).

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C104	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Rhode Island (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 12%	2.30	9.97	2.30 to 2.30	3039, 82	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C105	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Vermont (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 3%	0.77	2.16	0.77 to 0.77	3039, 80	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C106	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Tennessee (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	0.54	1.72	0.54 to 0.54	3039, 68	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C107	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Virginia (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	-0.66	0.52	-0.66 to -0.66	3039, 91	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C108	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Nevada (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	-0.66	0.52	-0.66 to -0.66	3039, 77	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C109	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Mississippi (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	0.39	1.47	0.39 to 0.39	3039, 77	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C110	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Kansas (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	0.38	1.46	0.38 to 0.38	3039, 77	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C111	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Montana (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	0.38	1.46	0.38 to 0.38	3039, 75	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C112	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to New York (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	0.38	1.46	0.38 to 0.38	3039, 93	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C113	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Indiana (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 2%	0.37	1.45	0.37 to 0.37	3039, 82	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C114	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Delaware (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	-0.20	0.82	-0.20 to -0.20	3039, 81	agrees; AUC gain 0.015 [0.003, 0.029]	yes

ID	What is checked	Risk, typical patient	Log-odds	OR	Range (p10–p90)	n	Test check	OK
C115	direction: state: state other than Indiana, Kansas, Mississippi, Montana, Nevada, New York, Rhode Island, Tennessee, Virginia, Vermont, Delaware, Florida to Florida (through this part) *shape: categorical; highest to lowest: Rhode Island, Vermont, Tennessee, Mississippi, Kansas, Montana …*	1% to 1%	-0.14	0.87	-0.14 to -0.14	3039, 78	agrees; AUC gain 0.015 [0.003, 0.029]	yes
C116	structure: its formula reproduces it on 4000 training patients (largest difference 1.11022e-16)		1e-16			4000	train only	yes

The network before any explanation

Figure 47 (next page, landscape) is the trained network as it is, with no parts, no splits and no inserted units: inputs at the bottom, hidden units by depth, the output at the top, every connection drawn, every input column its own node (one-hot levels included).

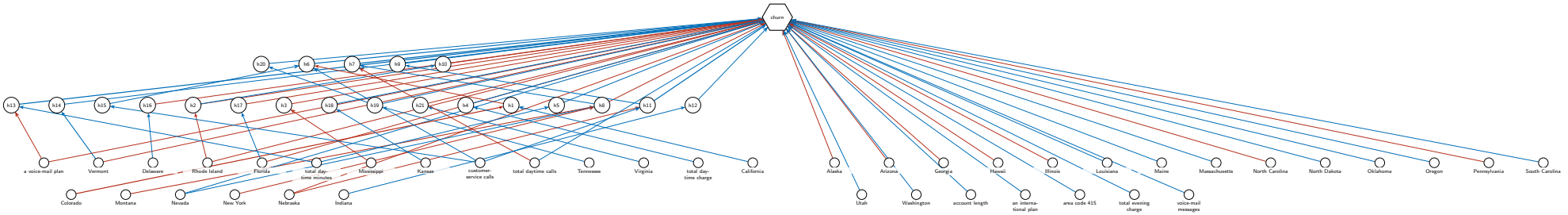


Figure 47: The network as trained, before any explanation, in full. Blue: positive weight (raises the score); red: negative weight (lowers it); thicker: larger weight. Orange outline and up arrow: raises the final prediction; teal and down arrow: lowers it; grey and double arrow: depends. Not used (no connection): Alabama, Iowa, Idaho, Kentucky, Arkansas, Maryland, Michigan, Minnesota, Missouri, New Hampshire, New Jersey, New Mexico, Ohio, South Dakota, Texas, Wisconsin, West Virginia, Wyoming, Connecticut, Washington DC, area code 408, area code 510, total evening minutes, total evening calls, total night-time minutes, total night-time calls, total night-time charge, total international minutes, total international calls, total international charge.